

24-25-SIM-Midterm Template

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Contents

1	Mobile Price Prediction Dataset	1
2	Load dataset and define factors	1

1 Mobile Price Prediction Dataset

Load mobiles.RData file in your current R or RStudio session. Selected numeric target is Price and let company be the qualitative target. Prepare your dataset to represent factors in a suitable way.

```
## null device
##          1
```

All questions account for 1 point (you have to answer all of them)

2 Load dataset and define factors

```
##          brand          company      Ratings      RAM
## Kechaoda A27 : 11 Samsung : 70 Min. :2.800 Min. : 0.000
## Easyfone Star: 8 Ikall : 61 1st Qu.:3.900 1st Qu.: 6.000
## Jivi R21Plus : 8 Kechaoda: 49 Median :4.100 Median : 6.000
## Lava 34 : 8 OPPO : 47 Mean :4.112 Mean : 6.116
## Lava A1 : 7 Micax : 42 3rd Qu.:4.400 3rd Qu.: 8.000
## MTR Ferrari : 7 Lava : 41 Max. :4.700 Max. :30.000
## (Other) :719 (Other) :458 NA's :23 NA's :7
## ROM Mobile_Size Primary_Cam Selfi_Cam
## Min. : 0.00 Min. : 2.00 Min. : 5.00 Min. : 0.00
## 1st Qu.: 32.00 1st Qu.: 4.50 1st Qu.:48.00 1st Qu.: 5.00
## Median : 64.00 Median : 4.77 Median :48.00 Median : 8.00
## Mean : 65.86 Mean : 5.59 Mean :48.25 Mean : 9.71
## 3rd Qu.: 64.00 3rd Qu.: 6.30 3rd Qu.:48.00 3rd Qu.:13.00
## Max. :256.00 Max. :44.00 Max. :64.00 Max. :61.00
## NA's :4 NA's :2 NA's :241
## Battery_Power Price
## Min. :1020 Min. : 539
## 1st Qu.:3000 1st Qu.: 990
## Median :3000 Median : 1699
## Mean :3268 Mean : 19071
## 3rd Qu.:3800 3rd Qu.: 18999
## Max. :6000 Max. :573000
##
```

All questions account for 1 point (you have to answer all of them)

1. Determine thresholds for mild and severe outliers in the target. How many mild and severe outliers are found? Indicate observation id's. Does the proportion of severe outliers depend on the company? *Remove those devices with prices above 200,000\$.*

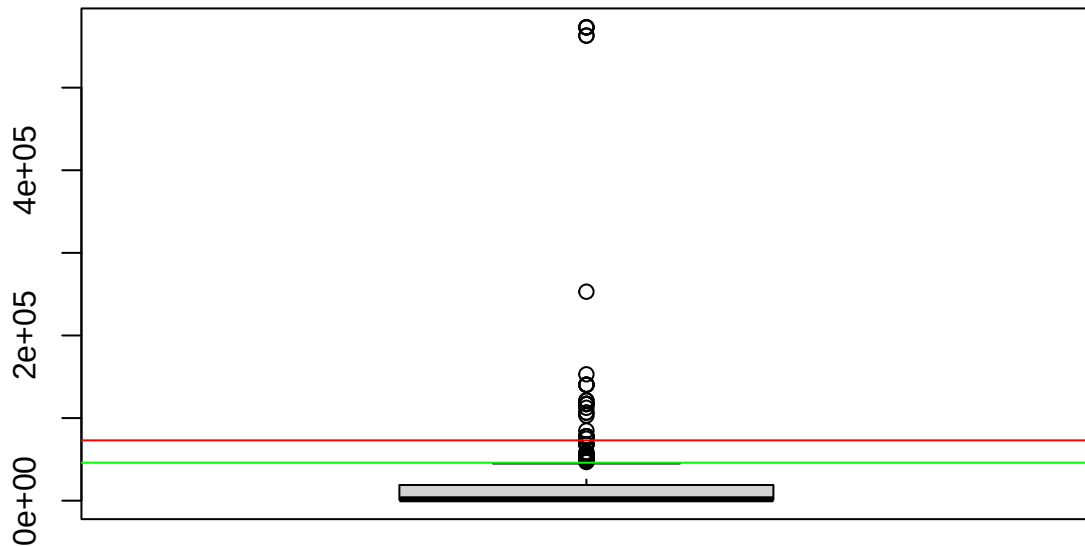
The thresholds can be found below and there are over 29 mild outliers and 7 are severe, however there is a set of incredible expensive devices which are over inflating the analysis. Performing a small analysis using FactoMineR condes, we can clearly see that there is a huge global association with both categorical variables brand (0.98 R2) and company (0.32 R2). Looking at particular individuals we can see that values of brand such as Iphone XR or samsung galaxy S10 have a bigger mean estimate value of price compared to other brands outside of apple. Additionally, we see that this variable has a moderate correlation with the Ratings, RAM, ROM and Battery_Power, while it has a weak negative correlation with Primary_cam.

Additionally, looking at the companies of the outliers we can clearly see that the severe ones are almost always either Apple or Samsung, followed by a few exceptions. We observe the similar trend in mild outliers.

```
library(FactoMineR)
Q3 <- quantile(df$Price, 0.75)
Q1 <- quantile(df$Price, 0.25)
IQR <- Q3 - Q1

mild_up <- Q3 + 1.5*IQR
mild_bot <- Q1 - 1.5*IQR
sev_up <- Q3 + 3*IQR
sev_bot <- Q1 - 3*IQR

boxplot(df$Price)
abline(h=mild_up, col="green")
abline(h=sev_up, col="red")
```



```

sev_ids <- which((df$Price > sev_up) | (df$Price < sev_bot))
mild_ids <- which(((df$Price > mild_up) & (df$Price < sev_up)) | ((df$Price < mild_bot) & (df$Price > s

df[sev_ids, "company"]

## [1] Apple Apple Apple Apple Apple Apple Apple Apple Apple
## [10] Apple Apple Apple Apple Apple Apple Apple Apple Apple
## [19] Apple Apple Apple LG Xiaomi Samsung Samsung Samsung Samsung
## [28] Samsung Samsung Samsung Samsung Samsung Vivo
## 33 Levels: Alcatel Apple Blacear BlackZone Easyfone Gee Hicell Honor ... Xiaomi

df[mild_ids, "company"]

## [1] Apple Apple Apple Apple Apple Apple Apple Apple Apple
## [10] Apple LG Mi Mi OnePlus OnePlus Samsung Samsung Samsung
## [19] Samsung Samsung Samsung Samsung Samsung Samsung Samsung Samsung Samsung
## [28] Samsung Samsung
## 33 Levels: Alcatel Apple Blacear BlackZone Easyfone Gee Hicell Honor ... Xiaomi

names(df)

## [1] "brand" "company" "Ratings" "RAM"
## [5] "ROM" "Mobile_Size" "Primary_Cam" "Selfi_Cam"
## [9] "Battery_Power" "Price"

condes(df, num.var = 10)

##
## Link between the variable and the continuous variables (R-square)

```

```
## =====
##               correlation      p.value
## Ratings      0.3698388 1.457731e-25
## RAM           0.2720602 2.218376e-14
## ROM           0.2492675 2.764084e-12
## Battery_Power 0.1725299 1.512232e-06
## Primary_Cam   -0.1673048 3.134071e-06
##
## Link between the variable and the categorical variable (1-way anova)
## =====
##               R2      p.value
## brand    0.9854507 1.138104e-243
## company  0.3223612 1.142405e-43
##
## Link between variable and the categories of the categorical variables
## =====
##                               Estimate      p.value
## brand=Apple iPhone XR ((PRODUCT)RED, 128 ) 550546.59 1.879445e-53
## brand=Samsung Galaxy S10 (Prism White, 512 ) 540546.59 2.669187e-51
## company=Apple                               122870.31 2.160103e-48
## brand=Apple iPhone XR (Yellow, 128 )        550546.59 1.495822e-25
## brand=Apple iPhone XR (Coral, 128 )         550546.59 1.495822e-25
## company=Samsung                             32809.86 4.188098e-06
## brand=Apple iPhone 11 Pro (Space Grey, 512 ) 117846.59 1.196228e-04
## brand=Apple iPhone 11 Pro Max (Gold, 64 )    94646.59 3.280686e-04
## brand=Apple iPhone 11 Pro (Silver, 512 )     117846.59 1.725334e-03
## brand=Vivo V17Pro (Glacier Ice White, 128 )   81545.93 7.180649e-03
## brand=Apple iPhone 11 Pro (Space Grey, 256 ) 98846.59 8.285047e-03
## brand=Redmi Note 8 (Cosmic Purple, 128 )     130546.59 1.454799e-02
## brand=Apple iPhone 11 Pro (Gold, 64 )        84146.59 2.387091e-02
## brand=Samsung Galaxy S20+ (Cloud Blue, 128 ) 55545.59 3.128075e-02
## company=Karbonn                             -13348.03 4.049181e-02
## company=Lava                                -13143.96 3.761086e-02
## company=Micax                               -13194.19 3.469355e-02
## company=Kechaoda                           -13868.38 1.721721e-02
## company=Ikall                               -13665.54 8.018546e-03

expns_ids <- which(df$Price > 200000)
df = df[-expns_ids, ]
```

2. Use an imputation method to address missing data in the dataset. Validate imputation results. *Retain all imputed values.*

Firstly, we can see in the correlation matrix that the correlations between variables are pretty weak, being Price ~ Ratings 0.36 the biggest among them, followed by battery_power and ratings. We will remove the ids to perform PCA imputation. After performing an imputation and comparing the quantiles before and after, we can see that in some extreme cases like Selfi_Cam, the values have slightly incremented but the distribution is still almost the same. Similarly, other variables that contain NA like RAM or ROM also have tiny differences but can be ultimately considered negligible, therefore we will proceed with the imputed data.

```
summary(df)
```

##	brand	company	Ratings	RAM
##	Kechaoda A27 : 11	Samsung : 68	Min. :2.800	Min. : 0.000
##	Easyfone Star: 8	Ikall : 61	1st Qu.:3.825	1st Qu.: 6.000

```
## Jivi R21Plus : 8 Kechaoda: 49 Median :4.100 Median : 6.000
## Lava 34 : 8 OPPO : 47 Mean :4.108 Mean : 6.077
## Lava A1 : 7 Micax : 42 3rd Qu.:4.400 3rd Qu.: 6.000
## MTR Ferrari : 7 Lava : 41 Max. :4.700 Max. :30.000
## (Other) :712 (Other) :453 NA's :23 NA's :7
## ROM Mobile_Size Primary_Cam Selfi_Cam
## Min. : 0.00 Min. : 2.000 Min. : 5.00 Min. : 0.000
## 1st Qu.: 32.00 1st Qu.: 4.500 1st Qu.:48.00 1st Qu.: 5.000
## Median : 56.00 Median : 4.700 Median :48.00 Median : 8.000
## Mean : 65.54 Mean : 5.585 Mean :48.32 Mean : 9.715
## 3rd Qu.: 64.00 3rd Qu.: 6.300 3rd Qu.:48.00 3rd Qu.:13.000
## Max. :256.00 Max. :44.000 Max. :64.00 Max. :61.000
## NA's :4 NA's :2 NA's :241
## Battery_Power Price
## Min. :1020 Min. : 539
## 1st Qu.:3000 1st Qu.: 990
## Median :3000 Median : 1699
## Mean :3266 Mean : 14422
## 3rd Qu.:3800 3rd Qu.: 18999
## Max. :6000 Max. :153000
##
```

```
library(corrplot)
```

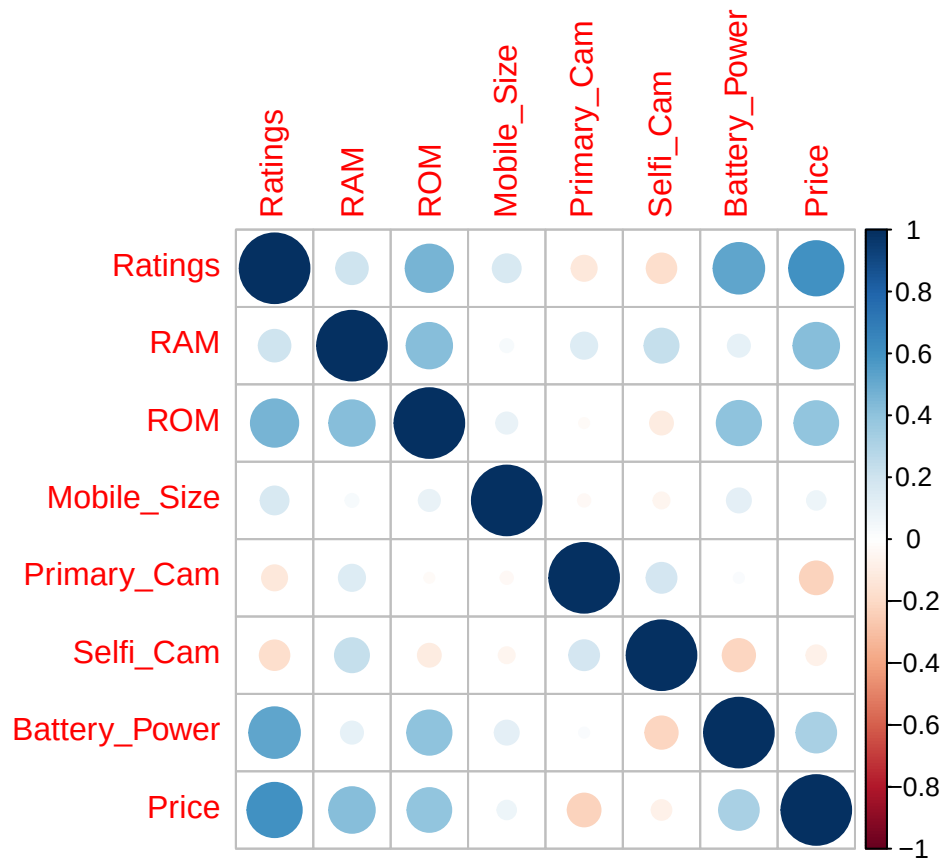
```
## corrplot 0.95 loaded
```

```
library(missMDA)
```

```
num_vars <- df[sapply(df, is.numeric)]
```

```
cat_vars <- df[!sapply(df, is.numeric)]
```

```
corrplot(cor(na.omit(num_vars)))
```



```
res_imp <- imputePCA(num_vars)
imp_vals <- res_imp$completeObs
```

```
names(num_vars)
```

```
## [1] "Ratings"      "RAM"          "ROM"          "Mobile_Size"
## [5] "Primary_Cam"  "Selfi_Cam"    "Battery_Power" "Price"
```

```
sapply(df, summary)
```

```
## $brand
##           Kechaoda A27
##                11
##           Easyfone Star
##                8
##           Jivi R21Plus
##                8
##           Lava 34
##                8
##           Lava A1
##                7
##           MTR Ferrari
##                7
##           I Kall K11
##                6
##           InFocus POWER 2
```

##	6
##	Inovu A9
##	6
##	Kechaoda A32
##	6
##	Nokia 105
##	6
##	Nokia 3310 DS
##	6
##	Snexian Guru 324
##	6
##	Snexian Guru 332
##	6
##	JIVI JV 12M
##	5
##	Karbons K24 Plus Pro
##	5
##	Kechaoda K33
##	5
##	Micax X749
##	5
##	Nokia 105 SS
##	5
##	Samsung Guru 1200
##	5
##	Alcatel 5V (Spectrum Blue, 32)
##	4
##	Apple iPhone 11 Pro Max (Gold, 64)
##	4
##	Blacear B5 Grip
##	4
##	Blacear I7 Trio
##	4
##	Easyfone Elite
##	4
##	Hicell C2 Dezire
##	4
##	I Kall K6610
##	4
##	InFocus Vibe 1
##	4
##	Jivi N3
##	4
##	Jivi N300 New
##	4
##	Jmax M40 Co of Two
##	4
##	Karbons K451 Power
##	4
##	Karbons KX21
##	4
##	Kechaoda K10
##	4
##	Kechaoda K111

##		4
##	Kechaoda K115	
##		4
##	Kechaoda K66 Plus	
##		4
##	Lava A1 Josh	
##		4
##	Lava A7	
##		4
##	LG Q7+ (Black, 64)	
##		4
##	Micax X381	
##		4
##	Micax X421	
##		4
##	Micax X742	
##		4
##	Moto One (White, 64)	
##		4
##	Nokia 105 ss	
##		4
##	OnePlus 7 Pro (Mirror Grey, 128)	
##		4
##	OnePlus 7 Pro (Nebula Blue, 256)	
##		4
##	OPPO A9 (Marble Green, 128)	
##		4
##	POCO F1 (Graphite Black, 256)	
##		4
##	POCO F1 (Steel Blue, 128)	
##		4
##	Samsung Galaxy A70s (Prism Crush Red, 128)	
##		4
##	Samsung Galaxy S10 Lite (Prism Blue, 512)	
##		4
##	Samsung Galaxy S20+ (Cloud Blue, 128)	
##		4
##	Samsung Galaxy S8 Plus (Midnight Black, 64)	
##		4
##	Apple iPhone 11 (Yellow, 64)	
##		3
##	Apple iPhone 11 Pro (Space Grey, 512)	
##		3
##	Apple iPhone 8 (PRODUCT)RED (Red, 256)	
##		3
##	Blacear i7 Trio	
##		3
##	BlackZone ATOM	
##		3
##	Easyfone Amico	
##		3
##	I Kall K28 Coo of Two Mobiles	
##		3
##	I Kall K33 New	

##		3
##	I Kall K3310	
##		3
##	I Kall K3312 Coo	
##		3
##	I kall k6300 NEW	
##		3
##	Itel It2171	
##		3
##	JIVI 12M	
##		3
##	Karbons K-Pebble	
##		3
##	Karbons K140 Pop	
##		3
##	Karbons K334	
##		3
##	Kechaoda K2	
##		3
##	Lava One	
##		3
##	LG Q Stylus (Black, 32)	
##		3
##	Micax X516	
##		3
##	OPPO R17 Pro (Emerald Green, 128)	
##		3
##	Redmi K20 (Glacier Blue, 128)	
##		3
##	Redmi Note 5 Pro (Gold, 64)	
##		3
##	Redmi Note 6 Pro (Rose Gold, 64)	
##		3
##	Samsung Galaxy A21s (White, 64)	
##		3
##	Samsung Galaxy A50 (Blue, 64)	
##		3
##	Samsung Galaxy A50 (White, 64)	
##		3
##	Samsung Galaxy S10 Lite (Prism Black, 512)	
##		3
##	Vivo S1 (Diamond Black, 128)	
##		3
##	Vivo Y91i (Black, 32)	
##		3
##	Vivo Y91i (Ocean Blue, 16)	
##		3
##	Apple iPhone 11 (Red, 64)	
##		2
##	Apple iPhone 11 Pro (Gold, 64)	
##		2
##	Apple iPhone 11 Pro (Silver, 512)	
##		2
##	Apple iPhone 11 Pro (Space Grey, 256)	

```

##                                     2
##      Apple iPhone 7 Plus (Gold, 32 )
##                                     2
##      Blacear B5 Nex
##                                     2
##      Blacear B5 Tork
##                                     2
##      Blacear C-99 Sultan
##                                     2
##      BlackZone M4
##                                     2
##      Easyfone Royale
##                                     2
##      Easyfone Udaan
##                                     2
##      Gee F103 Pro (Gold, 16 )
##                                     2
##      Gee F103 Pro (Grey, 16 )
##                                     2
##      Gee S11 Lite (Gold, 32 )
##                                     2
##      (Other)
##      377
##
## $company
##      Alcatel      Apple      Blacear BlackZone Easyfone      Gee      Hicell      Honor
##      7           34          17          6           19          7          6          12
##      Ikall      InFocus      Inovu      iQ00      Itel      JIVI      Jmax      Karbonn
##      61          14          8          8          22          28          6          39
##      Kechaoda    Lava          LG          Mi          Micax    Motorola    MTR      Nokia
##      49          41          11         8          42          5          16          34
##      OnePlus     OPPO      Realme     Salora     Samsung    Snexian     Ssky     Vivo
##      19          47          16          7          68          16          7          40
##      Xiaomi
##      41
##
## $Ratings
##      Min. 1st Qu.  Median      Mean 3rd Qu.  Max.  NA's
##      2.800  3.825  4.100  4.108  4.400  4.700  23
##
## $RAM
##      Min. 1st Qu.  Median      Mean 3rd Qu.  Max.  NA's
##      0.000  6.000  6.000  6.077  6.000 30.000  7
##
## $ROM
##      Min. 1st Qu.  Median      Mean 3rd Qu.  Max.  NA's
##      0.00  32.00  56.00  65.54  64.00 256.00  4
##
## $Mobile_Size
##      Min. 1st Qu.  Median      Mean 3rd Qu.  Max.  NA's
##      2.000  4.500  4.700  5.585  6.300 44.000  2
##
## $Primary_Cam
##      Min. 1st Qu.  Median      Mean 3rd Qu.  Max.

```

```
##      5.00   48.00   48.00   48.32   48.00   64.00
##
## $Selfi_Cam
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.      NA's
##      0.000   5.000   8.000   9.715  13.000  61.000      241
##
## $Battery_Power
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1020   3000   3000   3266   3800   6000
##
## $Price
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      539     990   1699   14422   18999  153000
```

```
na_positions <- is.na(num_vars)
num_vars[na_positions] <- imp_vals[na_positions]
df[sapply(df, is.numeric)] <- num_vars
sapply(df, summary)
```

```
## $brand
##
##      Kechaoda A27
##      11
##      Easyfone Star
##      8
##      Jivi R21Plus
##      8
##      Lava 34
##      8
##      Lava A1
##      7
##      MTR Ferrari
##      7
##      I Kall K11
##      6
##      InFocus POWER 2
##      6
##      Inovu A9
##      6
##      Kechaoda A32
##      6
##      Nokia 105
##      6
##      Nokia 3310 DS
##      6
##      Snexian Guru 324
##      6
##      Snexian Guru 332
##      6
##      JIVI JV 12M
##      5
##      Karbonn K24 Plus Pro
##      5
##      Kechaoda K33
##      5
##      Micax X749
```

##	5
##	Nokia 105 SS
##	5
##	Samsung Guru 1200
##	5
##	Alcatel 5V (Spectrum Blue, 32)
##	4
##	Apple iPhone 11 Pro Max (Gold, 64)
##	4
##	Blacear B5 Grip
##	4
##	Blacear I7 Trio
##	4
##	Easyfone Elite
##	4
##	Hicell C2 Dezire
##	4
##	I Kall K6610
##	4
##	InFocus Vibe 1
##	4
##	Jivi N3
##	4
##	Jivi N300 New
##	4
##	Jmax M40 Coo of Two
##	4
##	Karbons K451 Power
##	4
##	Karbons KX21
##	4
##	Kechaoda K10
##	4
##	Kechaoda K111
##	4
##	Kechaoda K115
##	4
##	Kechaoda K66 Plus
##	4
##	Lava A1 Josh
##	4
##	Lava A7
##	4
##	LG Q7+ (Black, 64)
##	4
##	Micax X381
##	4
##	Micax X421
##	4
##	Micax X742
##	4
##	Moto One (White, 64)
##	4
##	Nokia 105 ss

##		4
##	OnePlus 7 Pro (Mirror Grey, 128)	
##		4
##	OnePlus 7 Pro (Nebula Blue, 256)	
##		4
##	OPPO A9 (Marble Green, 128)	
##		4
##	POCO F1 (Graphite Black, 256)	
##		4
##	POCO F1 (Steel Blue, 128)	
##		4
##	Samsung Galaxy A70s (Prism Crush Red, 128)	
##		4
##	Samsung Galaxy S10 Lite (Prism Blue, 512)	
##		4
##	Samsung Galaxy S20+ (Cloud Blue, 128)	
##		4
##	Samsung Galaxy S8 Plus (Midnight Black, 64)	
##		4
##	Apple iPhone 11 (Yellow, 64)	
##		3
##	Apple iPhone 11 Pro (Space Grey, 512)	
##		3
##	Apple iPhone 8 (PRODUCT)RED (Red, 256)	
##		3
##	Blacear i7 Trio	
##		3
##	BlackZone ATOM	
##		3
##	Easyfone Amico	
##		3
##	I Kall K28 Coo of Two Mobiles	
##		3
##	I Kall K33 New	
##		3
##	I Kall K3310	
##		3
##	I Kall K3312 Coo	
##		3
##	I kall k6300 NEW	
##		3
##	Itel It2171	
##		3
##	JIVI 12M	
##		3
##	Karbons K-Pebble	
##		3
##	Karbons K140 Pop	
##		3
##	Karbons K334	
##		3
##	Kechaoda K2	
##		3
##	Lava One	

##		3
##	LG Q Stylus (Black, 32)	
##		3
##	Micax X516	
##		3
##	OPPO R17 Pro (Emerald Green, 128)	
##		3
##	Redmi K20 (Glacier Blue, 128)	
##		3
##	Redmi Note 5 Pro (Gold, 64)	
##		3
##	Redmi Note 6 Pro (Rose Gold, 64)	
##		3
##	Samsung Galaxy A21s (White, 64)	
##		3
##	Samsung Galaxy A50 (Blue, 64)	
##		3
##	Samsung Galaxy A50 (White, 64)	
##		3
##	Samsung Galaxy S10 Lite (Prism Black, 512)	
##		3
##	Vivo S1 (Diamond Black, 128)	
##		3
##	Vivo Y91i (Black, 32)	
##		3
##	Vivo Y91i (Ocean Blue, 16)	
##		3
##	Apple iPhone 11 (Red, 64)	
##		2
##	Apple iPhone 11 Pro (Gold, 64)	
##		2
##	Apple iPhone 11 Pro (Silver, 512)	
##		2
##	Apple iPhone 11 Pro (Space Grey, 256)	
##		2
##	Apple iPhone 7 Plus (Gold, 32)	
##		2
##	Blacear B5 Nex	
##		2
##	Blacear B5 Tork	
##		2
##	Blacear C-99 Sultan	
##		2
##	BlackZone M4	
##		2
##	Easyfone Royale	
##		2
##	Easyfone Udaan	
##		2
##	Gee F103 Pro (Gold, 16)	
##		2
##	Gee F103 Pro (Grey, 16)	
##		2
##	Gee S11 Lite (Gold, 32)	

```
##
##                2
##              (Other)
##              377
##
## $company
##   Alcatel      Apple   Blacear BlackZone Easyfone      Gee   Hicell      Honor
##        7        34       17        6        19        7        6        12
##   Ikall   InFocus    Inovu    iQ00    Itel    JIVI    Jmax   Karbonn
##       61        14        8        8        22       28        6        39
## Kechaoda   Lava      LG      Mi      Micax  Motorola    MTR    Nokia
##       49        41       11       8       42        5       16       34
##   OnePlus    OPPO    Realme   Salora   Samsung  Snexian    Ssky    Vivo
##       19        47       16        7        68       16        7       40
##   Xiaomi
##       41
##
```

```
## $Ratings
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  2.800  3.889   4.100   4.103  4.400   4.700
##
```

```
## $RAM
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000  6.000   6.000   6.085  6.909  30.000
##
```

```
## $ROM
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00  32.00   56.00   65.47  64.00  256.00
##
```

```
## $Mobile_Size
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  2.000  4.500   4.770   5.584  6.300  44.000
##
```

```
## $Primary_Cam
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5.00  48.00   48.00   48.32  48.00   64.00
##
```

```
## $Selfi_Cam
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    8.00   11.36   10.44  13.00   61.00
##
```

```
## $Battery_Power
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1020    3000    3000    3266    3800    6000
##
```

```
## $Price
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    539    990   1699   14422   18999  153000
##
```

```
summary(df)
```

```
##           brand           company           Ratings           RAM
## Kechaoda A27 : 11 Samsung : 68 Min. :2.800 Min. : 0.000
## Easyfone Star: 8 Ikall : 61 1st Qu.:3.889 1st Qu.: 6.000
## Jivi R21Plus : 8 Kechaoda: 49 Median :4.100 Median : 6.000
## Lava 34 : 8 OPPO : 47 Mean :4.103 Mean : 6.085
```

```
## Lava A1      : 7   Micax   : 42   3rd Qu.:4.400   3rd Qu.: 6.909
## MTR Ferrari : 7   Lava    : 41   Max.      :4.700   Max.      :30.000
## (Other)     :712  (Other) :453
##      ROM      Mobile_Size      Primary_Cam      Selfi_Cam
## Min.      : 0.00   Min.      : 2.000   Min.      : 5.00   Min.      : 0.00
## 1st Qu.: 32.00   1st Qu.: 4.500   1st Qu.:48.00   1st Qu.: 8.00
## Median : 56.00   Median : 4.770   Median :48.00   Median :11.36
## Mean    : 65.47   Mean    : 5.584   Mean    :48.32   Mean    :10.44
## 3rd Qu.: 64.00   3rd Qu.: 6.300   3rd Qu.:48.00   3rd Qu.:13.00
## Max.    :256.00   Max.    :44.000   Max.    :64.00   Max.    :61.00
##
## Battery_Power      Price
## Min.      :1020   Min.      : 539
## 1st Qu.:3000   1st Qu.: 990
## Median :3000   Median : 1699
## Mean    :3266   Mean    :14422
## 3rd Qu.:3800   3rd Qu.:18999
## Max.    :6000   Max.    :153000
##
```

3. Are there multivariate outliers in the dataset? Indicate how many and which at 99% confidence. Explain what they seem to share in common. Do not take any action.

There are 51 multivariant outliers and they all same to share a similar structure, the company and brand is Apple, the ratings are atleast 1 point bigger, and there is a huge difference in mean price.

```
library(chemometrics)
```

```
## Cargando paquete requerido: rpart
```

```
num_vars <- df[sapply(df, is.numeric)]
num_vars$Primary_Cam <- NULL
num_vars$Ratings <- NULL
num_vars$Mobile_Size <- NULL
num_vars$RAM <- NULL
num_vars
```

```
##      ROM Selfi_Cam Battery_Power      Price
## 1      8.00000    6.303301          2000    4999
## 2      8.00000    6.303301          2000    4999
## 3     32.00000   16.000000          3800   12999
## 4     32.00000    1.000000          3800    9790
## 5     32.00000    1.000000          3800    9790
## 6     32.00000    1.000000          3800    9790
## 7     32.00000    1.000000          3800    9790
## 8     64.00000   12.000000          5000   68300
## 9     64.00000   12.000000          3500   68300
## 10    64.00000   12.000000          3800   68300
## 11    64.00000   12.000000          3500   68300
## 12    64.00000   12.000000          3500   68300
## 13    64.00000   12.000000          3500  106600
## 14    64.00000   12.000000          3500  106600
## 15    64.00000   12.000000          5000  140300
## 16    64.00000   12.000000          3500  140300
## 17   256.00000   12.000000          3800  121300
## 18    64.00000   12.000000          3500  121300
```

## 19	128.00000	12.000000	5000	140300
## 20	64.00000	12.000000	3500	140300
## 21	64.00000	12.000000	3500	140300
## 22	64.00000	12.000000	3000	117100
## 23	64.00000	12.000000	3500	117100
## 24	64.00000	12.000000	3500	117100
## 25	64.00000	12.000000	3500	117100
## 26	64.00000	12.000000	3500	117100
## 27	64.00000	12.000000	3500	117100
## 28	64.00000	5.000000	3500	28999
## 29	64.00000	12.000000	3500	36999
## 30	64.00000	12.000000	3500	36999
## 31	64.00000	12.000000	3500	36999
## 32	64.00000	12.000000	3500	36999
## 33	256.00000	7.000000	5000	56999
## 34	64.00000	7.000000	3500	56999
## 35	64.00000	7.000000	3500	56999
## 36	64.00000	7.000000	3500	38999
## 37	64.00000	7.000000	5000	38999
## 38	64.00000	12.000000	3500	67820
## 39	64.00000	12.000000	3500	49900
## 44	64.00000	12.000000	3500	84900
## 45	64.00000	12.000000	3500	112450
## 46	32.00000	17.000000	1050	799
## 47	32.00000	14.000000	1050	799
## 48	32.00000	15.228022	1050	799
## 49	32.00000	15.000000	1050	799
## 50	64.00000	15.935492	1050	799
## 51	64.00000	15.935492	1050	799
## 52	64.00000	8.322034	3000	799
## 53	64.00000	8.322034	3000	799
## 54	64.00000	5.000000	2500	1079
## 55	64.00000	15.000000	2500	1079
## 56	32.00000	13.000000	1550	1699
## 57	24.00000	18.179426	1550	1690
## 58	32.00000	14.860728	1550	1699
## 59	32.00000	16.000000	1550	1699
## 60	24.00000	18.179426	1550	1690
## 61	32.00000	14.860728	1550	1699
## 62	24.00000	14.000000	1550	1690
## 69	64.00000	13.242185	3800	899
## 70	64.00000	23.000000	3800	899
## 71	64.00000	13.245401	3800	699
## 72	32.00000	11.886481	2500	705
## 73	32.00000	12.000000	2500	705
## 74	32.00000	8.000000	1100	685
## 81	32.00000	11.392566	3800	2599
## 82	32.00000	11.606452	3500	2599
## 83	32.00000	11.606452	3500	2599
## 84	32.00000	10.993735	3000	3149
## 85	32.00000	8.000000	3000	3149
## 86	32.00000	20.000000	3000	3149
## 87	32.00000	20.000000	3000	3149
## 88	32.00000	20.000000	3500	3399

## 89	32.00000	20.000000	3500	3399
## 90	32.00000	8.000000	3000	3999
## 91	32.00000	12.354746	4300	3999
## 92	32.00000	12.925109	3500	3999
## 93	32.00000	5.000000	3000	3999
## 94	32.00000	5.000000	5000	3999
## 95	32.00000	12.996404	3400	3999
## 96	32.00000	22.000000	2500	3999
## 97	32.00000	12.000000	2000	3999
## 98	32.00000	11.140525	3000	2100
## 99	32.00000	11.140525	3000	2100
## 107	64.00000	1.000000	4010	21499
## 108	16.00000	5.000000	4200	8990
## 109	16.00000	5.000000	4200	8990
## 110	16.00000	5.000000	4200	8798
## 111	16.00000	5.000000	2500	8798
## 112	32.00000	2.000000	3030	9999
## 113	32.00000	2.000000	3030	9999
## 128	32.00000	5.000000	1050	629
## 129	32.00000	13.547081	1050	629
## 130	32.00000	13.547081	1050	629
## 131	32.00000	13.547081	1050	629
## 132	64.00000	11.358625	3500	629
## 133	64.00000	11.358625	3500	629
## 134	128.00000	2.000000	3750	22999
## 135	128.00000	2.000000	3750	22999
## 136	128.00000	2.000000	3750	22999
## 137	128.00000	2.000000	3750	13699
## 138	64.00000	2.000000	3750	19999
## 139	32.00000	2.000000	3000	13999
## 140	32.00000	2.000000	3000	13999
## 141	128.00000	2.000000	3000	19999
## 142	64.00000	2.000000	3000	15999
## 143	128.00000	2.000000	3000	19999
## 144	128.00000	8.000000	3800	16999
## 145	64.00000	2.000000	3750	25999
## 147	64.00000	12.000000	3000	639
## 148	32.00000	11.509144	3000	1199
## 149	32.00000	8.000000	3000	639
## 150	64.00000	12.000000	4000	15999
## 151	64.00000	5.000000	3000	1099
## 152	64.00000	14.000000	3000	599
## 153	64.00000	12.034840	3000	599
## 154	64.00000	12.034840	3000	599
## 155	64.00000	10.822818	4700	599
## 156	64.00000	12.026799	3000	1099
## 157	64.00000	14.000000	3000	699
## 158	64.00000	4.000000	3000	1199
## 159	64.00000	8.000000	3000	639
## 160	64.00000	11.874569	3000	639
## 161	64.00000	8.000000	3000	1199
## 162	16.00000	5.000000	2500	4999
## 163	64.00000	11.714780	3000	649
## 164	32.00000	12.986432	3000	649

## 165	32.00000	8.000000	3000	649
## 166	64.00000	5.000000	3000	1099
## 167	64.00000	5.000000	3000	1199
## 168	64.00000	11.867171	3000	1099
## 169	64.00000	20.000000	3000	669
## 170	64.00000	11.874408	3000	649
## 171	64.00000	8.000000	3000	639
## 172	64.00000	8.000000	3500	639
## 173	64.00000	8.446773	3500	639
## 174	64.00000	8.436159	3500	1299
## 175	64.00000	8.437767	3500	1199
## 176	64.00000	5.000000	3500	1299
## 177	64.00000	12.000000	1500	749
## 178	64.00000	12.617809	1500	749
## 179	64.00000	8.000000	1500	749
## 180	64.00000	11.555153	3000	649
## 181	64.00000	11.555314	3000	639
## 182	64.00000	11.555314	3000	639
## 183	64.00000	12.000000	3000	1199
## 184	64.00000	20.000000	3000	1199
## 185	64.00000	8.000000	3000	749
## 186	64.00000	20.000000	3000	1299
## 187	64.00000	16.000000	3000	1199
## 188	64.00000	20.000000	3000	1199
## 189	64.00000	22.000000	3000	1199
## 190	64.00000	11.714780	3000	649
## 191	64.00000	11.714780	3000	649
## 192	64.00000	11.705936	3000	1199
## 193	64.00000	16.000000	3000	1199
## 194	64.00000	11.865563	3000	1199
## 195	32.00000	5.000000	3000	1499
## 196	32.00000	11.334358	3000	1499
## 197	32.00000	11.334358	3000	1499
## 198	64.00000	11.875212	3000	599
## 199	64.00000	5.000000	3000	599
## 200	64.00000	11.875212	3000	599
## 201	64.00000	11.875212	3000	599
## 202	64.00000	6.890797	3500	899
## 203	64.00000	6.890797	3500	899
## 204	64.00000	8.000000	3500	539
## 205	64.00000	12.000000	3000	1099
## 206	64.00000	21.000000	3000	1099
## 207	64.00000	11.714941	3000	639
## 209	32.00000	11.177630	3000	1640
## 210	32.00000	20.000000	3000	950
## 211	64.00000	14.251599	2500	1199
## 212	64.00000	13.000000	2500	1390
## 213	64.00000	14.251599	2500	1199
## 214	64.00000	14.251599	2500	1199
## 215	64.00000	14.248528	2500	1390
## 216	64.00000	14.248528	2500	1390
## 217	32.00000	15.000000	3000	999
## 218	32.00000	11.833239	3000	898
## 219	32.00000	11.833239	3000	898

## 220	32.00000	11.833239	3000	898
## 221	32.00000	11.833239	3000	898
## 222	32.00000	12.712131	3000	990
## 223	32.00000	12.119438	3000	785
## 224	32.00000	11.763765	3500	735
## 225	32.00000	11.958570	1200	945
## 226	32.00000	0.000000	1200	945
## 227	32.00000	20.000000	1200	945
## 228	32.00000	0.000000	1200	945
## 229	32.00000	20.000000	1200	945
## 230	32.00000	0.000000	1200	945
## 232	25.00000	13.000000	4440	44990
## 233	25.00000	13.000000	4440	44990
## 234	128.00000	13.000000	4440	34990
## 235	128.00000	13.000000	4440	34990
## 236	256.00000	13.000000	4440	37990
## 237	256.00000	13.000000	4440	37990
## 238	128.00000	13.000000	4440	34990
## 239	128.00000	13.000000	4440	34990
## 240	32.00000	13.000000	3500	899
## 241	32.00000	32.000000	3500	998
## 242	32.00000	8.000000	3500	948
## 243	32.00000	12.000000	3000	949
## 244	32.00000	11.193909	3000	949
## 245	32.00000	20.000000	3000	1111
## 246	32.00000	15.000000	3000	1099
## 247	32.00000	20.000000	3000	1199
## 248	32.00000	11.353471	2500	1099
## 249	32.00000	5.000000	2500	1225
## 250	32.00000	11.195549	3000	847
## 251	32.00000	13.000000	3000	847
## 252	32.00000	11.035889	3000	849
## 253	32.00000	20.000000	3000	850
## 254	32.00000	11.001298	3000	3000
## 255	32.00000	12.627666	1200	1199
## 256	32.00000	12.626476	1200	1273
## 257	32.00000	32.000000	1900	1570
## 258	32.00000	13.000000	1900	1570
## 259	32.00000	20.000000	1200	1499
## 260	8.00000	13.356733	1900	1680
## 261	4.00000	12.000000	3000	1540
## 262	32.00000	10.000000	3500	799
## 263	32.00000	8.605384	3500	799
## 264	32.00000	8.605384	3500	799
## 265	32.00000	0.000000	3000	1200
## 266	32.00000	15.816884	3000	799
## 267	0.00000	26.595871	3500	799
## 268	32.00000	15.460407	3500	799
## 269	32.00000	15.460407	3500	799
## 270	32.00000	15.460407	3500	799
## 271	64.00000	9.584423	3500	949
## 272	32.00000	20.000000	3500	949
## 273	32.00000	20.000000	3500	949
## 274	32.00000	20.000000	3500	949

## 275	32.00000	8.000000	3000	1049
## 276	32.00000	8.000000	3000	999
## 277	32.00000	8.000000	3000	999
## 278	32.00000	8.000000	3000	1049
## 279	16.00000	8.000000	3800	2299
## 280	16.00000	8.000000	3000	1399
## 281	16.00000	8.000000	3000	1399
## 282	16.00000	7.905715	3000	798
## 283	16.00000	2.000000	3000	869
## 284	16.00000	8.000000	3000	869
## 285	16.00000	8.000000	3000	798
## 286	16.00000	8.000000	3000	869
## 287	16.00000	8.000000	3000	798
## 288	16.00000	8.000000	3000	869
## 289	16.00000	8.000000	3000	798
## 290	32.00000	14.910375	1050	699
## 291	32.00000	14.910375	1050	699
## 292	32.00000	8.000000	1050	1299
## 293	32.00000	14.741098	1050	1299
## 294	32.00000	14.741098	1050	1299
## 295	32.00000	12.994362	3500	1299
## 296	32.00000	11.176826	3000	1690
## 297	10.00000	11.000000	3000	920
## 298	32.00000	11.355723	3000	813
## 299	32.00000	5.000000	3000	849
## 300	32.00000	11.355145	3000	849
## 301	32.00000	12.243925	1750	999
## 302	32.00000	11.352732	3000	999
## 303	32.00000	11.352732	3000	999
## 304	32.00000	11.644960	2800	1299
## 305	32.00000	11.644960	2800	1299
## 306	32.00000	11.644960	2800	1299
## 307	32.00000	11.643496	2800	1390
## 308	32.00000	12.000000	2800	1299
## 309	32.00000	12.000000	2800	1149
## 310	32.00000	12.529463	1750	927
## 311	32.00000	12.529463	1750	927
## 312	32.00000	12.000000	1750	927
## 313	32.00000	16.000000	3000	1299
## 314	32.00000	11.501565	3000	1349
## 315	32.00000	8.000000	3000	1349
## 316	32.00000	16.000000	3000	1349
## 317	32.00000	11.501565	3000	1349
## 318	32.00000	11.148304	3500	1149
## 319	32.00000	32.000000	1750	999
## 320	64.00000	12.803977	1750	1399
## 321	64.00000	12.803977	1750	1399
## 322	12.00000	13.000000	3000	1579
## 323	12.00000	5.000000	3000	1579
## 324	12.00000	14.862343	3000	1579
## 325	32.00000	11.354340	3000	899
## 326	32.00000	5.000000	3000	1330
## 327	32.00000	11.346762	3000	1049
## 328	32.00000	11.342243	3000	1330

## 329	32.00000	13.000000	3000	1330
## 330	32.00000	11.966559	1900	1349
## 331	32.00000	11.000000	1900	1399
## 332	32.00000	11.341937	3000	1349
## 333	32.00000	11.341937	3000	1349
## 334	4.00000	2.000000	2800	13298
## 335	32.00000	11.768235	4700	990
## 336	32.00000	12.623779	3500	990
## 337	32.00000	11.555314	5000	930
## 338	32.00000	5.000000	3500	980
## 339	32.00000	12.000000	3500	945
## 340	32.00000	15.000000	3500	970
## 341	32.00000	12.623940	3500	980
## 342	32.00000	23.000000	3500	930
## 343	32.00000	15.000000	3500	945
## 344	32.00000	22.000000	3500	990
## 345	32.00000	12.623779	3500	990
## 346	32.00000	11.526386	3000	930
## 347	32.00000	11.526386	3000	930
## 348	32.00000	14.000000	3000	930
## 349	32.00000	11.526386	3000	930
## 350	32.00000	10.956023	3800	930
## 351	32.00000	32.000000	3800	930
## 352	32.00000	8.000000	3000	975
## 353	32.00000	7.610136	3000	980
## 354	32.00000	7.610216	3000	975
## 355	32.00000	23.000000	3000	980
## 356	32.00000	11.033702	3000	985
## 357	32.00000	11.033702	3000	985
## 358	32.00000	11.033702	3000	985
## 359	32.00000	11.033702	3000	985
## 360	64.00000	10.784465	3600	1799
## 361	64.00000	10.784465	3600	1799
## 362	32.00000	11.403900	2500	935
## 363	32.00000	12.000000	2500	970
## 364	32.00000	11.403900	2500	935
## 365	32.00000	11.403337	2500	970
## 366	32.00000	12.000000	3800	980
## 367	32.00000	11.158536	3000	980
## 368	32.00000	11.000000	3000	900
## 369	32.00000	15.000000	3000	900
## 370	32.00000	11.354324	3000	900
## 371	32.00000	5.000000	3000	970
## 372	32.00000	15.000000	3000	935
## 373	32.00000	15.000000	3000	970
## 374	32.00000	15.000000	3000	940
## 375	32.00000	5.931947	3000	970
## 376	32.00000	20.000000	3700	1300
## 377	52.57125	12.584430	2500	1045
## 378	64.00000	11.827739	3500	1045
## 379	64.00000	10.911239	3000	985
## 380	32.00000	11.403096	2500	985
## 381	64.00000	10.911239	3000	985
## 382	32.00000	11.403096	2500	985

## 383	64.00000	10.402743	3700	1250
## 384	32.00000	14.000000	2500	1695
## 385	64.00000	32.000000	2500	1710
## 386	32.00000	13.376461	2500	1699
## 387	32.00000	13.376525	2500	1695
## 388	32.00000	13.376525	2500	1695
## 389	32.00000	13.376461	2500	1699
## 390	32.00000	11.000000	2500	1695
## 391	32.00000	13.376525	2500	1695
## 392	32.00000	13.058492	2500	1619
## 393	32.00000	13.058492	2500	1619
## 394	24.00000	14.314744	3000	1149
## 395	24.00000	12.859070	5000	3000
## 396	24.00000	13.960679	3500	999
## 397	24.00000	14.314744	3000	1149
## 398	3.00000	12.000000	3000	949
## 399	32.00000	5.000000	3000	1049
## 400	32.00000	12.000000	4700	1049
## 401	64.00000	11.087432	3500	1098
## 402	64.00000	16.000000	3000	1099
## 403	64.00000	22.000000	3500	1098
## 404	64.00000	11.443893	3000	1099
## 405	3.00000	10.687137	1750	1299
## 406	3.00000	10.687137	1750	1299
## 407	24.00000	11.777626	1750	1799
## 408	24.00000	11.780842	1750	1599
## 409	24.00000	12.000000	1750	1480
## 410	24.00000	8.000000	1750	1799
## 411	51.00000	12.826702	1750	1599
## 412	32.00000	12.000000	1750	1799
## 413	49.36356	14.667225	1750	1840
## 414	64.00000	14.611212	1750	1840
## 415	32.00000	13.425519	1750	1799
## 416	24.00000	14.318025	3500	799
## 417	25.00000	12.000000	3000	1369
## 418	32.00000	11.017054	3000	1699
## 419	32.00000	5.000000	3000	1699
## 420	32.00000	23.000000	3000	1599
## 421	32.00000	8.000000	3000	1599
## 422	32.00000	14.145281	1200	1599
## 423	16.00000	2.000000	2500	5049
## 424	16.00000	5.000000	3000	6999
## 427	128.00000	13.000000	3800	54990
## 428	32.00000	8.000000	3000	15899
## 429	32.00000	8.000000	3000	15899
## 430	32.00000	8.000000	3000	15899
## 431	32.00000	5.000000	3000	103000
## 432	64.00000	5.000000	3000	13999
## 433	64.00000	5.000000	3000	13999
## 434	64.00000	5.000000	3000	13999
## 435	64.00000	5.000000	3000	13999
## 436	128.00000	13.000000	4000	24999
## 437	128.00000	13.000000	3000	24999
## 446	256.00000	20.000000	4780	53990

## 447	128.00000	20.000000	4780	51987
## 448	64.00000	12.000000	3010	14990
## 449	128.00000	8.000000	4030	17398
## 450	128.00000	8.000000	4030	14990
## 451	128.00000	8.000000	4030	17190
## 452	128.00000	5.000000	3700	16999
## 453	128.00000	12.000000	3000	16999
## 455	32.00000	1.000000	2900	9990
## 456	32.00000	1.000000	2900	9990
## 457	40.00000	8.000000	3800	1799
## 458	32.00000	5.000000	1750	1110
## 459	32.00000	12.000000	1750	1110
## 460	32.00000	13.000000	3000	1350
## 461	40.00000	9.817496	3000	1080
## 462	40.00000	15.000000	3000	1080
## 463	32.00000	15.000000	3000	959
## 464	32.00000	15.000000	3000	995
## 465	32.00000	11.477550	3000	995
## 466	32.00000	5.000000	4700	959
## 467	32.00000	12.014002	4600	1399
## 468	32.00000	8.669718	3000	899
## 469	32.00000	11.458841	2800	1099
## 470	32.00000	5.000000	2800	982
## 471	32.00000	12.000000	2800	982
## 472	32.00000	11.458841	2800	1099
## 473	24.00000	8.000000	1750	984
## 474	24.00000	11.090719	1750	984
## 475	32.00000	12.201781	1750	1451
## 476	32.00000	12.201781	1750	1451
## 477	32.00000	12.201781	1750	1451
## 478	2.40000	11.889207	3000	1190
## 479	64.00000	14.967472	2800	1460
## 480	32.00000	23.000000	3000	1240
## 481	32.00000	8.853570	3000	1240
## 482	32.00000	11.000000	3800	1280
## 483	32.00000	15.000000	2500	1240
## 484	32.00000	18.000000	1750	1549
## 485	32.00000	61.000000	1750	1390
## 486	32.00000	5.000000	1750	1549
## 487	32.00000	13.000000	1750	1549
## 488	32.00000	22.000000	1750	1390
## 489	32.00000	5.000000	2500	1259
## 490	32.00000	15.000000	2500	1259
## 491	20.00000	9.737193	3700	1499
## 492	25.00000	22.448056	3000	1199
## 493	32.00000	8.692511	3000	1329
## 494	32.00000	8.692511	3000	1329
## 495	56.00000	10.000000	3000	1799
## 496	56.00000	13.000000	3000	1799
## 497	64.00000	2.000000	3000	14999
## 498	64.00000	2.000000	3000	14999
## 499	64.00000	2.000000	3000	14999
## 500	64.00000	2.000000	3000	14999
## 501	128.00000	8.000000	5000	17499

## 502	64.00000	5.000000	3000	649
## 503	64.00000	11.711701	3000	649
## 504	64.00000	11.450526	3800	1275
## 505	64.00000	8.000000	5000	1275
## 506	64.00000	12.020889	3000	1275
## 507	64.00000	18.000000	3000	1275
## 508	64.00000	18.000000	3000	1275
## 509	64.00000	18.000000	3000	1275
## 510	64.00000	12.020889	3000	1275
## 511	64.00000	6.000000	3000	650
## 512	64.00000	12.509839	3000	649
## 513	64.00000	12.829078	3000	650
## 514	32.00000	11.004231	3000	649
## 515	32.00000	16.000000	3000	649
## 516	64.00000	11.870525	3000	699
## 517	64.00000	32.000000	3000	749
## 524	4.00000	13.000000	3500	1599
## 525	4.00000	13.000000	3500	1575
## 526	4.00000	13.000000	3500	1575
## 527	4.00000	13.000000	3500	1599
## 528	4.00000	13.000000	3500	1575
## 529	4.00000	13.000000	3500	1599
## 530	4.00000	11.316230	3500	1149
## 531	4.00000	11.316230	3500	1149
## 532	4.00000	16.000000	4000	15000
## 533	4.00000	11.126836	3500	3000
## 534	4.00000	11.000000	3500	1349
## 535	4.00000	11.118512	3500	1349
## 536	4.00000	11.126836	3500	3000
## 537	4.00000	11.118512	3500	1349
## 538	4.00000	11.126836	3500	3000
## 539	4.00000	11.118512	3500	1349
## 540	4.00000	11.118512	3500	1349
## 541	4.00000	5.000000	3500	1349
## 542	4.00000	15.000000	3500	1349
## 543	4.00000	11.220030	1020	2135
## 544	4.00000	11.220030	1020	2135
## 545	4.00000	11.230242	1020	1500
## 546	32.00000	2.000000	3800	9989
## 547	32.00000	2.000000	3800	9499
## 548	64.00000	20.000000	1020	2450
## 549	32.00000	7.215205	3800	9999
## 550	64.00000	11.477352	1200	3580
## 551	64.00000	11.464182	1200	4399
## 552	64.00000	11.477352	1200	3580
## 553	64.00000	11.466144	1200	4277
## 554	64.00000	13.000000	1200	3580
## 555	64.00000	15.000000	1200	3580
## 556	64.00000	5.000000	3500	18099
## 557	4.00000	14.000000	3500	1269
## 558	25.00000	2.000000	3000	39999
## 559	25.00000	2.000000	3000	39999
## 560	128.00000	5.000000	3000	32999
## 561	256.00000	5.000000	3000	36999

##	562	256.00000	5.000000	3000	35999
##	563	128.00000	8.000000	3800	42990
##	564	128.00000	8.000000	3800	42990
##	565	128.00000	8.000000	3800	42990
##	566	128.00000	8.000000	3800	42990
##	567	256.00000	8.000000	3800	43999
##	568	256.00000	8.000000	3800	43999
##	569	256.00000	8.000000	3800	42990
##	570	256.00000	8.000000	3800	42990
##	571	128.00000	1.000000	3000	37319
##	572	128.00000	1.000000	3000	37319
##	573	256.00000	1.000000	3000	38050
##	574	256.00000	1.000000	3000	38050
##	575	256.00000	1.000000	4085	46970
##	576	256.00000	1.000000	4085	46969
##	577	32.00000	2.000000	4230	9990
##	578	32.00000	2.000000	4230	9990
##	579	32.00000	5.000000	3800	8490
##	580	64.00000	2.000000	4230	12490
##	581	32.00000	2.000000	4230	8990
##	582	32.00000	2.000000	4230	9490
##	583	32.00000	2.000000	4230	13744
##	584	32.00000	2.000000	4230	8990
##	585	64.00000	2.000000	4230	15656
##	586	64.00000	2.000000	4230	15656
##	587	32.00000	2.000000	4230	12990
##	588	32.00000	2.000000	4230	12990
##	589	16.00000	8.000000	3180	9490
##	590	128.00000	2.000000	3500	16490
##	591	128.00000	2.000000	4020	16490
##	592	128.00000	2.000000	4020	16490
##	593	128.00000	2.000000	3500	16490
##	594	128.00000	2.000000	4020	16490
##	595	128.00000	8.000000	5000	16990
##	596	128.00000	8.000000	5000	16990
##	597	128.00000	8.000000	5000	15990
##	598	128.00000	8.000000	5000	15990
##	599	128.00000	8.000000	5000	15990
##	600	128.00000	5.000000	3800	17990
##	601	128.00000	5.000000	4700	17990
##	602	64.00000	5.000000	3800	24990
##	603	128.00000	8.000000	3800	18990
##	604	64.00000	2.000000	3500	19990
##	605	64.00000	2.000000	3500	18990
##	606	64.00000	2.000000	3350	18990
##	607	64.00000	2.000000	3600	18989
##	608	64.00000	13.000000	3600	18989
##	609	128.00000	10.593879	3500	22999
##	610	128.00000	5.000000	3500	22999
##	611	128.00000	20.000000	3000	31590
##	612	128.00000	20.000000	3000	31590
##	613	128.00000	20.000000	3000	31590
##	614	256.00000	13.000000	4065	41990
##	615	256.00000	13.000000	4065	41990

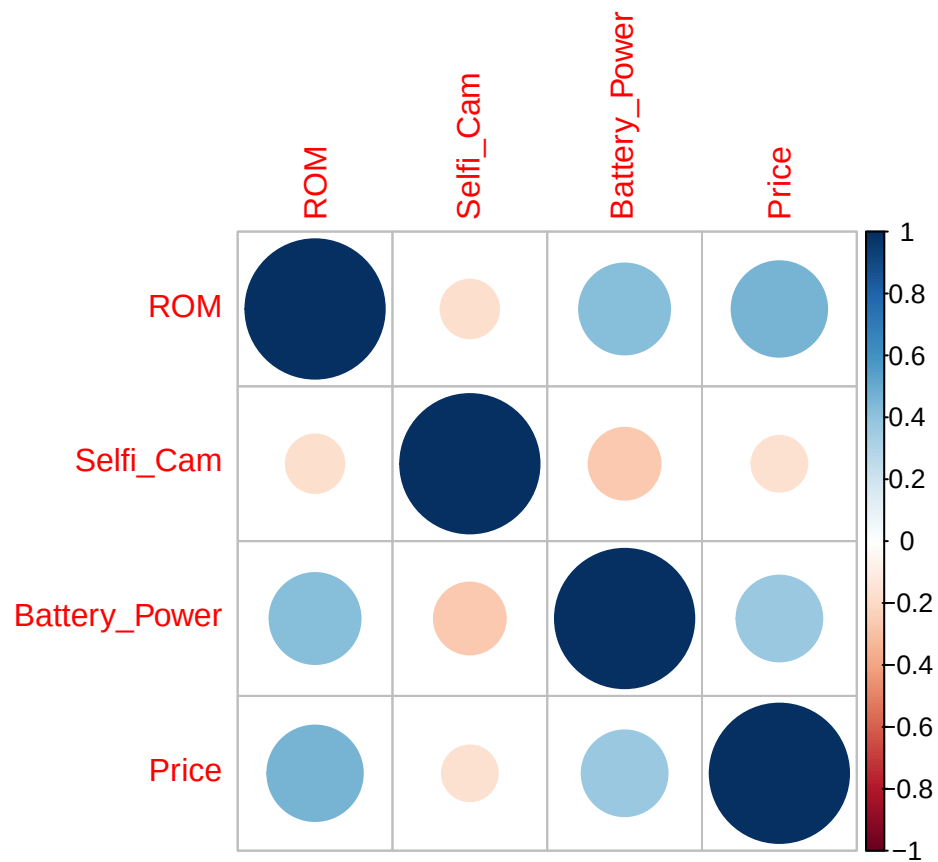
## 616	256.00000	8.000000	3800	38990
## 617	256.00000	8.000000	3800	38990
## 618	256.00000	8.000000	3800	38990
## 619	256.00000	8.000000	3800	38990
## 620	128.00000	13.000000	4025	29990
## 621	128.00000	13.000000	4025	29990
## 622	128.00000	13.000000	4025	29990
## 623	128.00000	13.000000	4025	29990
## 626	256.00000	5.000000	3800	22999
## 627	256.00000	5.000000	3800	22999
## 628	256.00000	5.000000	3800	22999
## 629	256.00000	5.000000	3800	22999
## 630	256.00000	5.000000	3800	22999
## 631	64.00000	5.000000	3800	21999
## 632	128.00000	15.000000	3800	18999
## 633	128.00000	5.000000	3800	18999
## 634	128.00000	5.000000	3800	18999
## 635	128.00000	5.000000	3800	18999
## 636	64.00000	8.000000	4700	17499
## 638	128.00000	2.000000	3500	15490
## 639	128.00000	2.000000	3500	15490
## 640	64.00000	5.000000	4045	15999
## 641	128.00000	8.000000	4700	16999
## 642	64.00000	12.000000	4700	17999
## 643	16.00000	2.000000	3800	6499
## 644	128.00000	8.000000	5000	11999
## 645	128.00000	8.000000	5000	11999
## 646	128.00000	8.000000	5000	11999
## 647	128.00000	13.000000	3800	31999
## 648	128.00000	13.000000	3800	31999
## 649	25.00000	13.000000	3800	35999
## 650	25.00000	13.000000	3800	35999
## 651	128.00000	13.000000	3800	31999
## 652	128.00000	13.000000	3800	31999
## 653	64.00000	13.000000	3800	29999
## 654	64.00000	2.000000	5000	9998
## 655	128.00000	13.000000	3800	24999
## 656	64.00000	13.000000	3800	21999
## 657	128.00000	13.000000	3800	24999
## 658	128.00000	13.000000	3800	24999
## 659	128.00000	13.000000	3800	24999
## 660	128.00000	13.000000	4030	24999
## 661	128.00000	13.000000	3800	24999
## 662	64.00000	13.000000	3800	21999
## 663	128.00000	13.000000	3800	24999
## 664	128.00000	13.000000	3000	24999
## 665	256.00000	13.000000	3800	29999
## 666	128.00000	13.000000	3000	24999
## 667	128.00000	13.000000	3800	24999
## 668	256.00000	13.000000	3800	29999
## 669	64.00000	5.000000	3800	13999
## 670	64.00000	5.000000	3800	13999
## 671	64.00000	5.000000	3800	13999
## 672	64.00000	5.000000	3800	17999

## 673	64.00000	5.000000	3000	14995
## 674	64.00000	5.000000	3000	16675
## 675	64.00000	5.000000	3000	15990
## 676	64.00000	5.000000	3000	16675
## 677	128.00000	5.000000	3800	17990
## 678	32.00000	5.000000	3800	9999
## 679	128.00000	1.000000	3800	153000
## 680	64.00000	8.000000	4700	16899
## 681	64.00000	8.000000	4700	16899
## 682	128.00000	8.000000	4550	17999
## 683	128.00000	8.000000	4700	16999
## 684	32.00000	13.654323	2800	899
## 685	32.00000	14.469361	1200	1299
## 686	32.00000	12.563835	4100	1149
## 687	32.00000	23.000000	4100	1149
## 688	32.00000	13.913009	2000	749
## 689	32.00000	12.000000	2000	749
## 690	32.00000	13.913009	2000	749
## 691	64.00000	8.000000	3800	14999
## 692	64.00000	8.000000	5000	18499
## 693	64.00000	8.000000	5000	18499
## 694	64.00000	8.000000	5000	16499
## 695	64.00000	8.000000	5000	18499
## 696	64.00000	8.000000	5000	18499
## 697	64.00000	5.000000	3800	18999
## 698	64.00000	5.000000	3800	18999
## 699	64.00000	5.000000	3800	18999
## 700	64.00000	15.000000	3800	18999
## 701	64.00000	5.000000	3800	18999
## 702	64.00000	5.000000	3800	18999
## 703	128.00000	8.000000	3800	18570
## 704	128.00000	8.000000	3800	18570
## 705	128.00000	12.000000	3800	25250
## 706	128.00000	5.000000	4700	29999
## 707	128.00000	5.000000	4700	29999
## 708	128.00000	5.000000	4700	29999
## 709	128.00000	5.000000	4700	29999
## 710	128.00000	5.000000	4700	25900
## 711	128.00000	12.000000	4700	32999
## 712	128.00000	12.000000	4700	32999
## 713	128.00000	8.000000	3000	40900
## 714	128.00000	8.000000	3000	52962
## 715	128.00000	5.000000	3000	27999
## 716	128.00000	5.000000	3000	27999
## 717	32.00000	2.000000	3800	9939
## 718	32.00000	2.000000	3800	9987
## 719	128.00000	8.000000	6000	19900
## 720	128.00000	8.000000	6000	19900
## 721	128.00000	8.000000	6000	19900
## 722	128.00000	8.000000	6000	19900
## 723	256.00000	12.000000	3500	75000
## 724	128.00000	12.000000	4700	39999
## 725	128.00000	12.000000	4700	39999
## 726	128.00000	12.000000	4700	39999

## 727	128.00000	12.000000	4700	39999
## 730	128.00000	12.000000	4700	47999
## 731	128.00000	12.000000	4700	47999
## 732	128.00000	12.000000	4700	47999
## 733	128.00000	12.000000	4700	47999
## 734	128.00000	12.000000	4700	47999
## 735	128.00000	12.000000	4700	47999
## 736	128.00000	12.000000	4700	47999
## 737	128.00000	12.000000	4700	42999
## 738	128.00000	12.000000	4700	42999
## 739	128.00000	12.000000	3100	59000
## 740	128.00000	1.000000	4700	77999
## 741	128.00000	1.000000	4700	77999
## 742	128.00000	1.000000	4700	77999
## 743	128.00000	1.000000	4700	77999
## 744	128.00000	1.000000	4700	77999
## 745	128.00000	1.000000	4700	77999
## 746	64.00000	8.000000	3500	53990
## 747	64.00000	8.000000	3500	53990
## 748	64.00000	8.000000	3500	53990
## 749	64.00000	8.000000	3500	53990
## 750	64.00000	8.000000	3500	53990
## 751	64.00000	12.610530	3500	1100
## 752	64.00000	8.000000	3500	1200
## 753	64.00000	12.608182	3500	1346
## 754	64.00000	13.000000	3500	1100
## 755	64.00000	12.608182	3500	1346
## 756	56.48521	10.184989	3800	1790
## 757	52.18490	11.297939	3500	1549
## 758	64.00000	12.968204	3500	1625
## 759	22.00000	9.746044	3000	2025
## 760	2.00000	10.145882	3000	2025
## 761	32.00000	8.000000	1080	649
## 762	32.00000	8.000000	1080	649
## 763	32.00000	10.095187	1080	649
## 764	32.00000	10.095187	1080	649
## 765	32.00000	13.149976	3000	649
## 766	32.00000	13.149976	3000	649
## 767	32.00000	15.000000	3000	649
## 768	32.00000	13.149976	3000	649
## 769	32.00000	13.149976	3000	649
## 770	32.00000	12.000000	3000	649
## 771	32.00000	32.000000	3000	649
## 772	32.00000	13.000000	3000	649
## 773	32.00000	13.149976	3000	649
## 774	32.00000	13.149976	3000	649
## 775	32.00000	21.000000	3000	649
## 776	32.00000	32.000000	3000	649
## 777	35.00000	15.056109	1050	849
## 778	35.00000	15.056109	1050	849
## 779	64.00000	13.237220	3000	749
## 780	64.00000	13.237220	3000	749
## 781	35.00000	23.000000	2800	949
## 782	32.00000	13.519311	3000	749

## 783	32.00000	8.000000	2800	1125
## 791	128.00000	5.000000	3800	29990
## 792	128.00000	5.000000	3800	29990
## 793	128.00000	8.000000	4700	16990
## 794	128.00000	8.000000	4700	16990
## 795	128.00000	8.000000	4700	19990
## 796	64.00000	8.000000	4700	18999
## 797	64.00000	8.000000	4700	18990
## 798	32.00000	8.000000	5000	10990
## 799	32.00000	8.000000	5000	10990
## 800	128.00000	32.000000	3000	21099
## 801	128.00000	8.000000	4700	24990
## 802	128.00000	8.000000	4700	24990
## 804	128.00000	13.000000	4100	29499
## 805	128.00000	13.000000	4100	29499
## 806	128.00000	13.000000	4100	29499
## 807	128.00000	13.000000	4100	29499
## 808	256.00000	8.000000	4700	31990
## 809	256.00000	8.000000	4700	31990
## 810	256.00000	8.000000	4700	31990
## 811	256.00000	8.000000	4700	31990
## 812	32.00000	2.000000	3000	18990
## 813	64.00000	2.000000	3225	22221
## 814	64.00000	5.000000	3260	23990
## 815	64.00000	5.000000	3260	23990
## 816	128.00000	5.000000	3200	36990
## 817	64.00000	2.000000	5000	12649
## 818	64.00000	2.000000	5000	12900
## 819	64.00000	2.000000	5000	12900
## 820	64.00000	2.000000	5000	12900
## 821	128.00000	2.000000	5000	13900
## 822	128.00000	8.000000	5000	16299
## 823	32.00000	1.000000	3000	14990
## 824	32.00000	5.000000	3260	13990
## 825	32.00000	5.000000	3260	13950
## 826	32.00000	5.000000	4030	8996
## 827	32.00000	6.734403	4030	10169
## 828	32.00000	5.000000	4030	8995
## 829	16.00000	5.000000	4030	9350
## 830	16.00000	5.000000	4030	9350
## 831	16.00000	5.000000	4030	9350

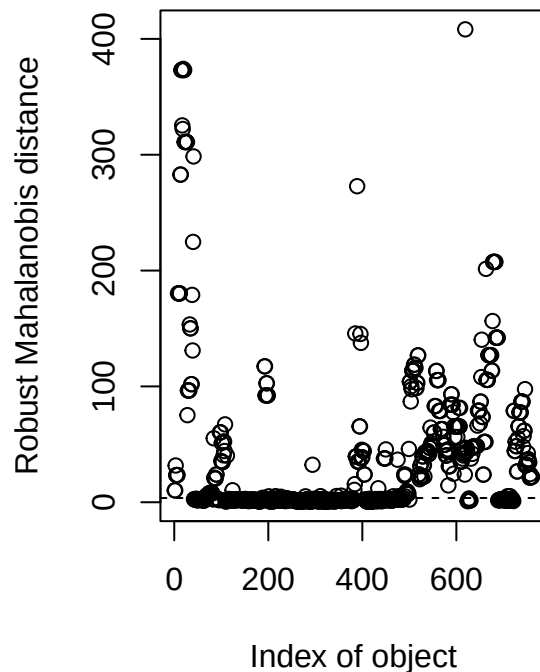
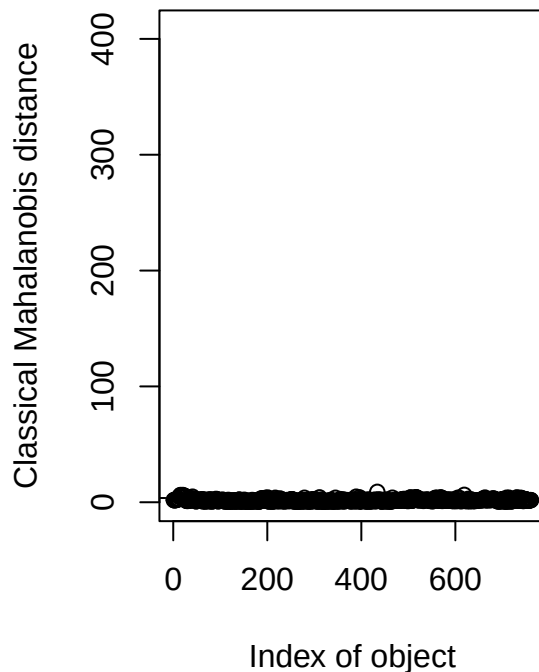
```
corrplot(cor(num_vars))
```



```
names(num_vars)
```

```
## [1] "ROM"          "Selfi_Cam"    "Battery_Power" "Price"
```

```
res_mout <- Moutlier(num_vars, quantile=0.99)
```



```
plot(x=res_mout$md, y=res_mout$rd, main="Classic Mahalanobis vs Robust")
abline(v=res_mout$cutoff, col="red")
abline(h=res_mout$cutoff, col="red")
```

```
mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))
mout_ids
```

```
## 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 45 236 237 241 257
## 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 41 196 197 201 217
## 319 351 385 431 446 485 517 561 562 573 574 575 576 614 615 616 617 618 619 626
## 279 311 345 389 396 434 466 504 505 516 517 518 519 557 558 559 560 561 562 567
## 627 628 629 630 665 668 679 723 771 776 800
## 568 569 570 571 605 608 619 663 709 714 731
```

```
summary(df[mout_ids, ])
```

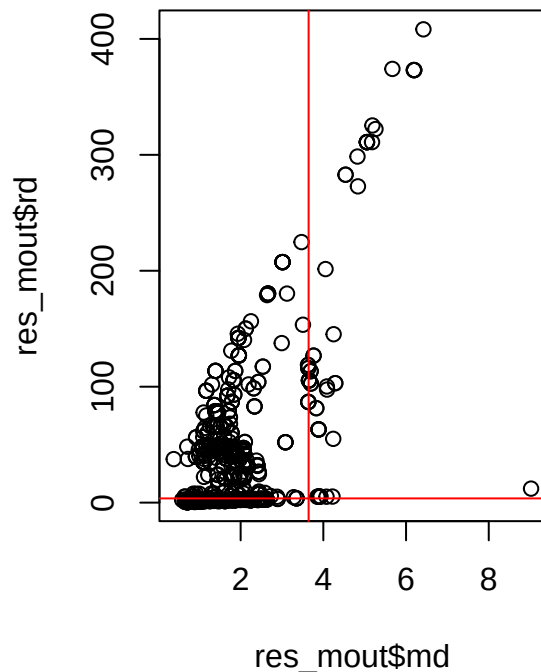
```
##          brand          company      Ratings
## Apple iPhone 11 Pro Max (Gold, 64 ) : 4 Apple :16 Min. :3.772
## POCO F1 (Graphite Black, 256 )      : 4 Xiaomi : 8 1st Qu.:4.450
## Apple iPhone 11 Pro (Space Grey, 512): 3 OnePlus: 6 Median :4.500
## Apple iPhone 11 Pro (Gold, 64 )      : 2 OPPO : 6 Mean :4.463
## Apple iPhone 11 Pro (Silver, 512 )    : 2 iQOO : 2 3rd Qu.:4.700
## Apple iPhone 11 Pro (Space Grey, 256 ): 2 Itel : 2 Max. :4.700
## (Other)                               :34 (Other):11
##      RAM      ROM      Mobile_Size      Primary_Cam
## Min. : 3.000 Min. : 32.0 Min. :4.40 Min. :26.00
## 1st Qu.: 8.000 1st Qu.: 64.0 1st Qu.:5.80 1st Qu.:35.00
```

```
## Median : 8.000 Median :128.0 Median :6.30 Median :48.00
## Mean : 7.706 Mean :153.1 Mean :5.96 Mean :42.51
## 3rd Qu.: 8.000 3rd Qu.:256.0 3rd Qu.:6.50 3rd Qu.:48.00
## Max. :12.000 Max. :256.0 Max. :6.67 Max. :64.00
##
## Selfi_Cam Battery_Power Price
## Min. : 1.00 Min. :1750 Min. : 649
## 1st Qu.: 6.50 1st Qu.:3000 1st Qu.: 22999
## Median :12.00 Median :3500 Median : 38990
## Mean :14.27 Mean :3531 Mean : 61216
## 3rd Qu.:13.00 3rd Qu.:3800 3rd Qu.:117100
## Max. :61.00 Max. :5000 Max. :153000
##
```

```
summary(df[-mout_ids,])
```

```
## brand company Ratings RAM
## Kechaoda A27 : 11 Samsung : 67 Min. :2.800 Min. : 0.000
## Easyfone Star: 8 Ikall : 61 1st Qu.:3.800 1st Qu.: 6.000
## Jivi R21Plus : 8 Kechaoda: 48 Median :4.100 Median : 6.000
## Lava 34 : 7 Micax : 41 Mean :4.077 Mean : 5.969
## Lava A1 : 7 OPPO : 41 3rd Qu.:4.400 3rd Qu.: 6.000
## MTR Ferrari : 7 Lava : 40 Max. :4.700 Max. :30.000
## (Other) :662 (Other) :412
## ROM Mobile_Size Primary_Cam Selfi_Cam
## Min. : 0.00 Min. : 2.000 Min. : 5.00 Min. : 0.00
## 1st Qu.: 32.00 1st Qu.: 4.500 1st Qu.:48.00 1st Qu.: 8.00
## Median : 32.00 Median : 4.580 Median :48.00 Median :11.35
## Mean : 59.18 Mean : 5.557 Mean :48.74 Mean :10.17
## 3rd Qu.: 64.00 3rd Qu.: 6.260 3rd Qu.:48.00 3rd Qu.:13.00
## Max. :256.00 Max. :44.000 Max. :64.00 Max. :26.60
##
## Battery_Power Price
## Min. :1020 Min. : 539
## 1st Qu.:3000 1st Qu.: 985
## Median :3000 Median : 1599
## Mean :3247 Mean :11061
## 3rd Qu.:3800 3rd Qu.:16997
## Max. :6000 Max. :84900
##
```

Classic Mahalanobis vs Robust



4. Analyze the profile of the numeric target (Price) using a suitable profiling method. A detailed explanation of the procedure outcome is requested.

Firsly, we see that Ratings and Price variable have a quite high correlation, followed by ROM, Battery power and RAM. There is also some weak inverse relation with Primary_cam and Selfi_Cam. It is already said in previous sections but the brand and company categorical values have a big global association with this variable. Delving deeper into companies, we can see that Apple has a huge estimate mean value of +73847.527 and other companies like samsung (19836) and OnePlus (28178) have estimated lower values. There is also the lower side of the spectrum such as Oppo(8366) and Xiaomi(10928).

```
names(df)
```

```
## [1] "brand"          "company"        "Ratings"        "RAM"
## [5] "ROM"            "Mobile_Size"    "Primary_Cam"    "Selfi_Cam"
## [9] "Battery_Power" "Price"
```

```
condes(df, num.var=10)
```

```
##
## Link between the variable and the continuous variables (R-square)
## =====
## correlation      p.value
## Ratings          0.60886306 2.182606e-78
## ROM               0.46920063 6.407444e-43
## Battery_Power     0.37910346 2.029491e-27
## RAM               0.26536623 9.898356e-14
## Mobile_Size       0.09436139 9.197899e-03
## Selfi_Cam         -0.16037295 8.763960e-06
```

```

## Primary_Cam    -0.24008707 1.938241e-11
##
## Link between the variable and the categorical variable (1-way anova)
## =====
##              R2          p.value
## brand    0.9992093  0.000000e+00
## company  0.7154880  3.630803e-175
##
## Link between variable and the categories of the categorical variables
## =====
##                                     Estimate      p.value
## company=Apple                      73847.527 2.170245e-97
## brand=Apple iPhone 11 Pro (Space Grey, 512 ) 123859.025 1.346542e-21
## brand=Apple iPhone 11 Pro Max (Gold, 64 ) 100659.025 3.073130e-19
## brand=Apple iPhone 11 Pro (Silver, 512 ) 123859.025 1.261919e-14
## company=Samsung                      19836.865 1.609913e-11
## brand=Apple iPhone 11 Pro (Space Grey, 256 ) 104859.025 7.533914e-11
## brand=Redmi Note 8 (Cosmic Purple, 128 ) 136559.025 2.668887e-09
## brand=Apple iPhone 11 Pro (Gold, 64 ) 90159.025 2.219226e-08
## brand=Samsung Galaxy S20+ (Cloud Blue, 128 ) 61558.025 4.734745e-08
## company=OnePlus                      28178.796 5.921471e-07
## brand=Apple iPhone 11 Pro Max (Space Grey, 64 ) 100659.025 1.155334e-05
## brand=Apple iPhone 11 Pro Max (Silver, 64 ) 100659.025 1.155334e-05
## brand=Apple iPhone XS (Silver, 512 ) 96009.025 2.853424e-05
## brand=Apple iPhone 11 (Yellow, 64 ) 51859.025 6.692239e-05
## brand=LG Q6 (Black, 32 ) 86559.025 1.584705e-04
## brand=Samsung Galaxy S8 Plus (Midnight Black, 64 ) 37549.025 7.295107e-04
## brand=Apple iPhone 11 (Red, 64 ) 51859.025 1.164289e-03
## brand=Apple iPhone 8 (PRODUCT)RED (Red, 256 ) 40558.025 1.663853e-03
## brand=Apple iPhone XS (Silver, 256 ) 68459.025 2.696887e-03
## company=iQOO                      25544.585 3.970311e-03
## brand=Samsung Galaxy S10 Lite (Prism Blue, 512 ) 31558.025 4.187123e-03
## brand=Samsung Galaxy S20+ (Cosmic Black, 128 ) 61558.025 6.829546e-03
## brand=Samsung Galaxy S20+ (Cosmic Gray, 128 ) 61558.025 6.829546e-03
## brand=Samsung Galaxy Note 10 (Aura Black, 256 ) 58559.025 9.969990e-03
## company=Xiaomi                      10928.707 1.002333e-02
## brand=OnePlus 7 Pro (Nebula Blue, 256 ) 27053.525 1.321788e-02
## brand=Samsung Galaxy S10 Lite (Prism Black, 512 ) 31558.025 1.325947e-02
## brand=OnePlus 7 Pro (Mirror Grey, 128 ) 26549.025 1.490515e-02
## brand=Apple iPhone 8 Plus (Gold, 256 ) 51379.025 2.318741e-02
## company=OPPO                      8366.968 4.592661e-02
## company=LG                      15639.222 4.837622e-02
## company=Easyfone                    -9359.467 3.763167e-02
## company=InFocus                    -11556.986 3.307526e-02
## company=MTR                        -11763.040 2.048247e-02
## company=Blacear                    -11494.473 1.913862e-02
## company=Snexian                    -12046.415 1.796938e-02
## company=Itel                       -11459.051 7.620331e-03
## company=Nokia                      -9009.121 6.460807e-03
## company=JIVI                       -11722.236 2.032280e-03
## company=Karbonn                    -11164.799 4.309928e-04
## company=Lava                       -10960.732 3.742864e-04
## company=Micax                      -11010.962 2.969023e-04
## company=Kechaoda                   -11685.149 3.467090e-05

```

```
## company=Ikall
```

```
-11482.316 4.335934e-06
```

5. Analyze the profile of the qualitative target (company) using a suitable method. A detailed explanation of the procedure outcome is requested. Focus on Ratings only in the explanation of numeric details.

Looking the results we can clearly see that company and brand have a perfect association on the chi-squared test, therefore the variables are not independent as it would be naturally expected. The qualitative variables which this factor has the highest global associations are mainly Ratings and Price, followed by the rest. Looking at the ratings we can clearly see a pattern that more expensive and known (Most importantly!) companies like Apple/Honor/Xiaomi perform better than the average rating (4.1), and less known companies as Hicell, Ikall, Inovu...

```
names(df)
```

```
## [1] "brand"          "company"        "Ratings"        "RAM"
## [5] "ROM"            "Mobile_Size"    "Primary_Cam"    "Selfi_Cam"
## [9] "Battery_Power" "Price"
```

```
res_cat <- catdes(df, num.var=2)
```

```
res_cat$test.chi2
```

```
##      p.value    df
## brand      0 12032
```

6. Is there variance homogeneity in the Price target groups defined by company classes? Indicate the 4 companies having the largest Price dispersion.

Looking at the inferential normality assessment method Shapiro test we can completely discard that the distribution of price is normal, therefore fligner test will be suited better for this purpose.

Both using fligner test and performing visual analytics with boxplots we can clearly see that there is no variance homogeneity in this sample at all, with a tiny p-value of 2.2e-16. We can also see that there are companies with no variance at all, and others with huge ones: Apple, LG, Samsung and Xiaomi in that descending order.

```
shapiro.test(df$Price)
```

```
##
## Shapiro-Wilk normality test
##
## data:  df$Price
## W = 0.62379, p-value < 2.2e-16
```

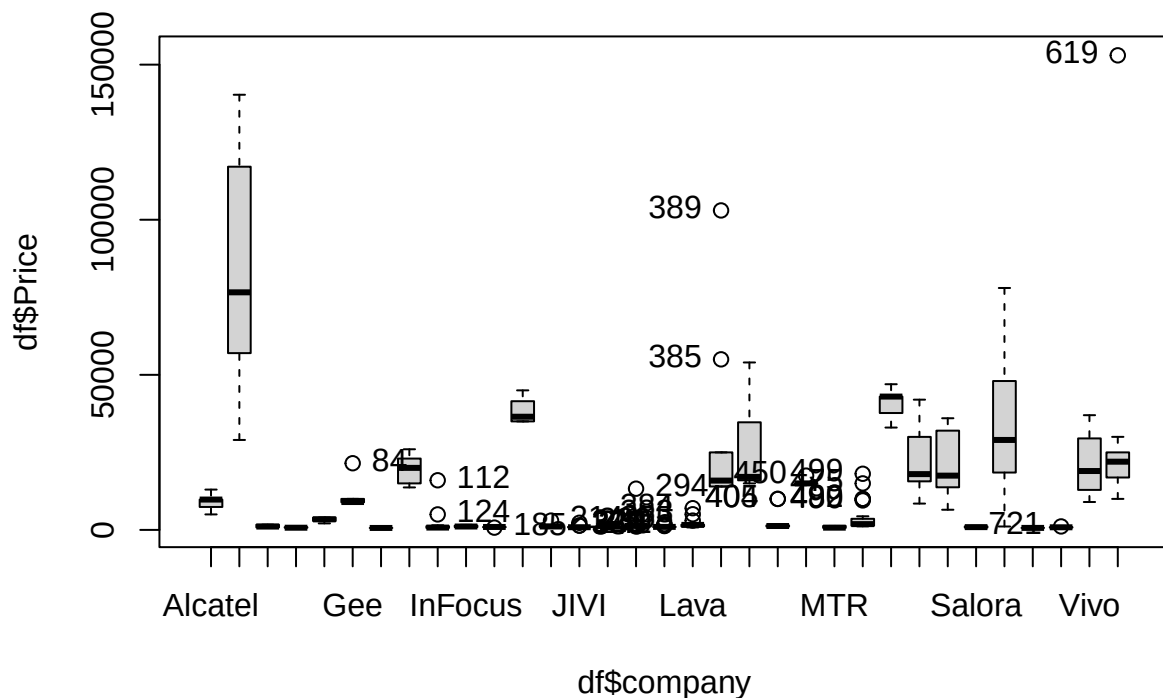
```
fligner.test(df$Price, df$company)
```

```
##
## Fligner-Killeen test of homogeneity of variances
##
## data:  df$Price and df$company
## Fligner-Killeen:med chi-squared = 519.05, df = 32, p-value < 2.2e-16
```

```
library(car)
```

```
## Cargando paquete requerido: carData
```

```
Boxplot(df$Price ~ df$company)
```



```
## [1] "84" "112" "124" "185" "214" "239" "240" "241" "294" "320" "321" "336"
## [13] "343" "355" "383" "384" "385" "389" "404" "405" "450" "475" "489" "490"
## [25] "492" "499" "721" "619"
```

```
sort(tapply(df$Price, df$company, var))
```

```
##      Hicell      Snexian      Inovu      BlackZone      Ssky      Kechaoda
## 0.000000e+00 0.000000e+00 7.512500e+03 1.077347e+04 1.922514e+04 3.189507e+04
##      Salora      InFocus      Jmax      JIVI      MTR      Blacear
## 5.392857e+04 5.851119e+04 9.600000e+04 9.614578e+04 9.805412e+04 1.894987e+05
##      Itel      Easyfone      Lava      Motorola      Micax      Karbonn
## 2.271533e+05 4.638481e+05 1.158186e+06 1.250000e+06 3.589982e+06 3.793633e+06
##      Ikall      Alcatel      OnePlus      Nokia      Honor      iQ00
## 4.065298e+06 8.400480e+06 1.468113e+07 1.654324e+07 1.734992e+07 1.907143e+07
##      Gee      Vivo      OPPO      Realme      Mi      Xiaomi
## 2.166834e+07 6.996209e+07 9.501242e+07 1.038399e+08 2.876224e+08 4.519800e+08
##      Samsung      LG      Apple
## 5.089615e+08 7.593655e+08 1.441589e+09
```

```
sort(tapply(df$Price, df$company, sd))
```

```
##      Hicell      Snexian      Inovu      BlackZone      Ssky      Kechaoda
## 0.000000 0.000000 86.67468 103.79531 138.65476 178.59192
##      Salora      InFocus      Jmax      JIVI      MTR      Blacear
## 232.22526 241.89086 309.83867 310.07383 313.13594 435.31447
##      Itel      Easyfone      Lava      Motorola      Micax      Karbonn
## 476.60601 681.06391 1076.19030 1118.03399 1894.72489 1947.72508
##      Ikall      Alcatel      OnePlus      Nokia      Honor      iQ00
```

```
## 2016.25840 2898.35821 3831.59637 4067.33874 4165.32403 4367.08468
##      Gee      Vivo      OPPO      Realme      Mi      Xiaomi
## 4654.92660 8364.33418 9747.43171 10190.18855 16959.43530 21259.82141
##      Samsung      LG      Apple
## 22560.17496 27556.58708 37968.26435
```

7. Mean Price can be considered to be the equal across groups defined by company factor? Use a two.sided test at 1% significance level and indicate the confidence interval for the particular case of Apple devices population mean. Indicate whether equal variances and normal distribution of Price hypothesis hold in the population.

Given that the distribution of price is not normal, we need to use a non-parametric test as Kruskal. Similarly, the mean of the different groups are extremely unlikely to be equal, with a tiny pvalue too. Therefore we reject it. We cannot use the equal variances assumption, therefore we will need to perform a wilcox.test instead of a t.test:

The given interval of the mean is (68060, 104300), which seems absolutely likely looking at the sample mean being 86542.

```
?kruskal.test
kruskal.test(df$Price, df$company)

##
## Kruskal-Wallis rank sum test
##
## data: df$Price and df$company
## Kruskal-Wallis chi-squared = 642.35, df = 32, p-value < 2.2e-16

?wilcox.test
apples <- df[which(df$company == "Apple"), ]
mean(apples$Price)

## [1] 86542.94

wilcox.test(apples$Price, paired=F, conf.level=0.99, alternative="two.sided", conf.int=T)

## Warning in wilcox.test.default(apples$Price, paired = F, conf.level = 0.99, :
## cannot compute exact p-value with ties

## Warning in wilcox.test.default(apples$Price, paired = F, conf.level = 0.99, :
## cannot compute exact confidence interval with ties

##
## Wilcoxon signed rank test with continuity correction
##
## data: apples$Price
## V = 595, p-value = 3.653e-07
## alternative hypothesis: true location is not equal to 0
## 99 percent confidence interval:
## 68060 104300
## sample estimates:
## (pseudo)median
## 87450
```

8. State and test one.sided hypothesis to assess whether Price is less for Xiaomi than the rest at 1% significance level. Indicate and justify a 95% two-sided confidence interval for Xiaomi population mean.

Using a one-sided wilcox.test we can see that the true mean of Xiaomi is extremely likely to be greater than the whole population mean, therefore we reject that xiaomi mean is less. Also looking at the sample mean we

can clearly see its way above. Using a two-sided 95% confidence test we get that the following confidence interval: (18999, 21999), which is still above the global mean 14k.

```
sort(tapply(df$Price, df$company, mean))
```

```
##      Hicell      Snexian BlackZone      Ssky      Inovu      MTR      Salora
## 629.0000 649.0000 765.3333 859.8571 898.7500 932.3750 963.2857
##      JIVI      Kechaoda      Jmax      InFocus      Blacear      Ikall      Itel
## 973.1786 1010.2653 1099.0000 1138.4286 1200.9412 1213.0984 1236.3636
##      Karbonn      Micax      Lava      Easyfone      Nokia      Alcatel      Gee
## 1530.6154 1684.4524 1734.6829 3335.9474 3686.2941 8879.5714 11010.4286
##      Motorola      Honor      Vivo      OPPO      Realme      Xiaomi      Mi
## 15499.0000 19140.6667 19945.5750 21062.3830 22154.1250 23624.1220 25567.8750
##      LG      Samsung      iQOO      OnePlus      Apple
## 28334.6364 32532.2794 38240.0000 40874.2105 86542.9412
```

```
global_mean <- mean(df$Price); global_mean
```

```
## [1] 14422.38
```

```
xiaomi <- df[which(df$company == "Xiaomi"), ]
```

```
wilcox.test(xiaomi$Price, mu=global_mean, paired=F, alternative="greater", conf.level=0.99)
```

```
## Warning in wilcox.test.default(xiaomi$Price, mu = global_mean, paired = F, :
## cannot compute exact p-value with ties
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: xiaomi$Price
```

```
## V = 824, p-value = 1.665e-07
```

```
## alternative hypothesis: true location is greater than 14422.38
```

```
wilcox.test(xiaomi$Price, mu=global_mean, paired=F, conf.level=0.95, conf.int=T)
```

```
## Warning in wilcox.test.default(xiaomi$Price, mu = global_mean, paired = F, :
## cannot compute exact p-value with ties
```

```
## Warning in wilcox.test.default(xiaomi$Price, mu = global_mean, paired = F, :
## cannot compute exact confidence interval with ties
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: xiaomi$Price
```

```
## V = 824, p-value = 3.33e-07
```

```
## alternative hypothesis: true location is not equal to 14422.38
```

```
## 95 percent confidence interval:
```

```
## 18999 21999
```

```
## sample estimates:
```

```
## (pseudo)median
```

```
## 20837
```

```
pairwise.wilcox.test(df$Price, df$company, conf.level=0.99)
```

```
## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties
```

```
## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties
```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties
## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties
## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties
## Warning in wilcox.test.default(xi, xj, paired = paired, ...): cannot compute
## exact p-value with ties

##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: df$Price and df$company
##
##      Alcatel Apple  Blacear BlackZone Easyfone Gee      Hicell Honor
## Apple      0.01168 -        -        -        -        -        -        -
## Blacear     0.03087 3.1e-06 -        -        -        -        -        -
## BlackZone  0.39479 0.02928 1.00000 -        -        -        -        -
## Easyfone    0.02699 8.9e-07 9.6e-05 0.05654 -        -        -        -
## Gee         1.00000 0.01169 0.03151 0.43053 0.02735 -        -        -
## Hicell      0.27490 0.02890 0.05810 0.37398 0.05248 0.30158 -        -
## Honor       0.08609 0.00013 0.00190 0.15365 0.00122 0.57502 0.12928 -
## Ikall       0.00977 3.7e-13 0.90354 1.00000 3.7e-07 0.00839 0.57614 2.5e-05
## InFocus    0.05912 2.8e-05 1.00000 0.66570 0.00042 0.06084 0.09881 0.00542
## Inovu      0.16542 0.00439 1.00000 0.96622 0.01292 0.17731 0.17714 0.04639
## iQ00       0.20661 0.10995 0.01730 0.30882 0.01373 0.21842 0.23055 0.05232
## Itel       0.02566 1.7e-07 1.00000 0.25350 3.8e-05 0.02583 0.05321 0.00080
## JIVI       0.01494 7.6e-09 1.00000 1.00000 3.9e-06 0.01504 0.03327 0.00026
## Jmax       0.36244 0.02920 1.00000 1.00000 0.05581 0.39216 0.32865 0.14667
## Karbonn    0.02167 1.1e-10 1.00000 0.05643 2.4e-06 0.01952 0.02562 8.2e-05
## Kechaoda   0.00688 5.8e-12 1.00000 0.02064 8.7e-08 0.00688 0.02051 3.8e-05
## Lava       0.01426 5.9e-11 1.00000 0.03059 1.5e-05 0.00909 0.02477 7.2e-05
## LG         0.10000 0.00691 0.00310 0.17634 0.00209 0.58442 0.14140 1.00000
## Mi         0.22137 0.03083 0.01784 0.33499 0.01407 1.00000 0.25411 1.00000
## Micax      0.03066 4.4e-11 1.00000 0.02735 8.1e-06 0.02164 0.02464 6.8e-05
## Motorola   0.54509 0.07765 0.13026 0.78367 0.12292 1.00000 0.40733 1.00000
## MTR        0.03637 6.2e-06 1.00000 1.00000 0.00015 0.03738 0.06590 0.00259
## Nokia     0.26454 6.4e-10 0.12928 0.02943 1.00000 0.20643 0.02905 0.00037
## OnePlus    0.03059 0.02082 0.00011 0.06506 4.9e-05 0.03083 0.06016 0.00139
## OPPO      0.11037 2.0e-10 5.7e-07 0.02228 1.2e-07 0.47453 0.02218 1.00000
## Realme     0.23343 1.5e-05 0.00032 0.08987 0.00017 0.78620 0.08180 1.00000
## Salora     0.28382 0.01169 1.00000 1.00000 0.02735 0.30158 0.29584 0.08881
## Samsung    0.36311 2.6e-07 2.7e-06 0.01617 0.00106 0.98404 0.01615 1.00000
## Snexian    0.00180 3.8e-06 6.9e-05 0.00283 4.4e-05 0.00190 0.00209 0.00033
## Ssky       0.28382 0.01169 1.00000 1.00000 0.02735 0.30158 0.29584 0.08881
## Vivo       0.22200 2.0e-10 1.3e-06 0.02583 3.3e-07 0.25930 0.02566 1.00000
## Xiaomi     0.01078 9.2e-10 1.1e-06 0.02380 2.6e-07 0.08881 0.02368 1.00000
##
##      Ikall  InFocus Inovu  iQ00  Itel  JIVI  Jmax  Karbonn
## Apple    -      -      -      -      -      -      -      -
## Blacear   -      -      -      -      -      -      -      -
## BlackZone -      -      -      -      -      -      -      -
## Easyfone  -      -      -      -      -      -      -      -
## Gee       -      -      -      -      -      -      -      -
## Hicell    -      -      -      -      -      -      -      -

```

## Honor	-	-	-	-	-	-	-	-
## Icall	-	-	-	-	-	-	-	-
## InFocus	1.00000	-	-	-	-	-	-	-
## Inovu	1.00000	1.00000	-	-	-	-	-	-
## iQ00	0.00160	0.03303	0.12126	-	-	-	-	-
## Itel	0.36311	1.00000	0.79704	0.01215	-	-	-	-
## JIVI	1.00000	0.72641	1.00000	0.00629	0.26454	-	-	-
## Jmax	1.00000	1.00000	1.00000	0.28837	1.00000	1.00000	-	-
## Karbonn	0.00061	1.00000	0.33051	0.00347	1.00000	0.00542	1.00000	-
## Kechaoda	1.00000	1.00000	1.00000	0.00232	1.00000	0.83334	1.00000	0.01238
## Lava	4.1e-07	0.10807	0.00645	0.00326	0.06751	3.9e-06	1.00000	0.02735
## LG	9.8e-05	0.00812	0.05581	1.00000	0.00150	0.00054	0.16582	0.00020
## Mi	0.00190	0.03411	0.12928	1.00000	0.01236	0.00635	0.31037	0.00348
## Micax	0.00024	1.00000	0.00626	0.00321	1.00000	8.0e-05	1.00000	1.00000
## Motorola	0.06658	0.21603	0.33499	0.43292	0.12815	0.08881	0.66570	0.06864
## MTR	1.00000	1.00000	1.00000	0.02119	1.00000	1.00000	1.00000	0.03087
## Nokia	7.1e-10	0.00869	0.00442	0.00449	0.00076	3.4e-07	0.12920	0.00013
## OnePlus	2.4e-08	0.00048	0.01504	1.00000	2.0e-05	3.3e-06	0.06375	3.8e-07
## OPPO	7.4e-16	7.4e-06	0.00261	0.06530	1.4e-08	2.8e-10	0.02228	1.9e-12
## Realme	6.0e-07	0.00123	0.02368	0.12420	8.4e-05	1.8e-05	0.08881	4.9e-06
## Salora	1.00000	1.00000	1.00000	0.21640	1.00000	1.00000	1.00000	0.86820
## Samsung	< 2e-16	1.4e-05	0.00154	1.00000	2.9e-08	3.3e-11	0.03559	3.0e-12
## Snexian	1.00000	0.00018	0.00106	0.00124	2.9e-05	8.6e-06	0.00261	2.1e-06
## Ssky	1.00000	0.60253	1.00000	0.21640	0.85163	1.00000	1.00000	0.12880
## Vivo	2.9e-14	1.4e-05	0.00341	0.00557	4.8e-08	1.4e-09	0.02583	2.5e-11
## Xiaomi	9.4e-15	1.2e-05	0.00303	0.00767	3.6e-08	1.0e-09	0.02378	7.8e-12
##	Kechaoda	Lava	LG	Mi	Micax	Motorola	MTR	Nokia
## Apple	-	-	-	-	-	-	-	-
## Blacear	-	-	-	-	-	-	-	-
## BlackZone	-	-	-	-	-	-	-	-
## Easyfone	-	-	-	-	-	-	-	-
## Gee	-	-	-	-	-	-	-	-
## Hicell	-	-	-	-	-	-	-	-
## Honor	-	-	-	-	-	-	-	-
## Icall	-	-	-	-	-	-	-	-
## InFocus	-	-	-	-	-	-	-	-
## Inovu	-	-	-	-	-	-	-	-
## iQ00	-	-	-	-	-	-	-	-
## Itel	-	-	-	-	-	-	-	-
## JIVI	-	-	-	-	-	-	-	-
## Jmax	-	-	-	-	-	-	-	-
## Karbonn	-	-	-	-	-	-	-	-
## Kechaoda	-	-	-	-	-	-	-	-
## Lava	5.5e-09	-	-	-	-	-	-	-
## LG	0.00010	0.00018	-	-	-	-	-	-
## Mi	0.00232	0.00327	1.00000	-	-	-	-	-
## Micax	1.2e-06	0.36528	0.00017	0.00321	-	-	-	-
## Motorola	0.05810	0.06751	1.00000	1.00000	0.06750	-	-	-
## MTR	1.00000	0.00291	0.00408	0.02177	0.25483	0.14690	-	-
## Nokia	9.2e-11	1.00000	0.00124	0.01277	0.00077	0.23931	7.0e-05	-
## OnePlus	8.9e-08	2.8e-07	0.55914	1.00000	2.5e-07	0.13923	0.00017	9.4e-07
## OPPO	1.6e-14	4.0e-13	1.00000	1.00000	5.8e-13	1.00000	1.3e-06	4.6e-10
## Realme	1.0e-06	2.9e-06	1.00000	1.00000	2.9e-06	1.00000	0.00048	4.2e-05
## Salora	1.00000	0.22137	0.10371	0.23055	0.89244	0.58442	1.00000	0.02228

```
## Samsung < 2e-16 6.5e-11 1.00000 1.00000 1.3e-12 1.00000 1.2e-06 1.2e-07
## Snexian 7.9e-07 1.8e-06 0.00044 0.00132 1.6e-06 0.00390 0.02449 3.9e-06
## Ssky 0.79408 0.03303 0.10371 0.23055 0.04585 0.58442 1.00000 0.01169
## Vivo 3.1e-13 4.9e-12 1.00000 1.00000 7.1e-12 1.00000 2.9e-06 4.1e-09
## Xiaomi 1.9e-13 3.1e-12 1.00000 1.00000 2.2e-12 1.00000 2.3e-06 3.5e-10
## OnePlus OPPO Realme Salora Samsung Snexian Ssky Vivo
## Apple - - - - - - - -
## Blacear - - - - - - - -
## BlackZone - - - - - - - -
## Easyfone - - - - - - - -
## Gee - - - - - - - -
## Hicell - - - - - - - -
## Honor - - - - - - - -
## Ikall - - - - - - - -
## InFocus - - - - - - - -
## Inovu - - - - - - - -
## iQ00 - - - - - - - -
## Itel - - - - - - - -
## JIVI - - - - - - - -
## Jmax - - - - - - - -
## Karbonn - - - - - - - -
## Kechaoda - - - - - - - -
## Lava - - - - - - - -
## LG - - - - - - - -
## Mi - - - - - - - -
## Micax - - - - - - - -
## Motorola - - - - - - - -
## MTR - - - - - - - -
## Nokia - - - - - - - -
## OnePlus - - - - - - - -
## OPPO 6.2e-06 - - - - - - -
## Realme 0.00031 1.00000 - - - - - -
## Salora 0.03083 0.00768 0.04639 - - - - -
## Samsung 1.00000 0.21151 1.00000 0.00847 - - - -
## Snexian 5.1e-05 1.0e-06 0.00011 0.00188 2.4e-07 - - - -
## Ssky 0.03083 0.00768 0.04639 1.00000 0.00586 0.00188 - - -
## Vivo 4.1e-07 1.00000 1.00000 0.00956 0.40010 2.0e-06 0.00956 -
## Xiaomi 1.6e-06 1.00000 1.00000 0.00854 1.00000 1.6e-06 0.00854 1.00000
##
## P value adjustment method: holm
```

9. The standard deviation of Price in the Xiaomi company should not exceed 25,000\$. For the sample in the Xiaomi devices in your dataset, calculate the standard deviation of Price assuming that normal assumption holds. State and include any assumptions needed to test at the 1% significance level whether population standard deviation is larger than 1 in the xiaomi group. Figure out the 99% upper threshold for Price in the xiaomi population standard deviation.

Assuming normality, we can make use of `varTest`, which at any given confidence level we cannot reject the null hypothesis that the standard deviation should be equal than 25000. *The sample standard deviation is actually 21259.82*, and the upper bound of the confidence interval at 99% confidence (in terms of standard deviation and not variance) is 28560. The lower bound given by the other one-sided test is 16848.12

```
sort(tapply(df$Price, df$company, sd))
```

```
##      Hicell      Snexian      Inovu      BlackZone      Ssky      Kechaoda
```

```
##      0.00000      0.00000      86.67468      103.79531      138.65476      178.59192
##      Salora      InFocus      Jmax      JIVI      MTR      Blacear
##      232.22526      241.89086      309.83867      310.07383      313.13594      435.31447
##      Itel      Easyfone      Lava      Motorola      Micax      Karbonn
##      476.60601      681.06391      1076.19030      1118.03399      1894.72489      1947.72508
##      Ikall      Alcatel      OnePlus      Nokia      Honor      iQOO
##      2016.25840      2898.35821      3831.59637      4067.33874      4165.32403      4367.08468
##      Gee      Vivo      OPPO      Realme      Mi      Xiaomi
##      4654.92660      8364.33418      9747.43171      10190.18855      16959.43530      21259.82141
##      Samsung      LG      Apple
##      22560.17496      27556.58708      37968.26435
```

```
sugma <- 25000 * 25000
library(EnvStats)
```

```
##
## Adjuntando el paquete: 'EnvStats'
## The following object is masked from 'package:car':
##
##      qqPlot
## The following objects are masked from 'package:stats':
##
##      predict, predict.lm
```

```
varTest(xiaomi$Price, conf.level=0.95, sigma.squared=sugma)
```

```
##
## Results of Hypothesis Test
## -----
##
## Null Hypothesis:          variance = 6.25e+08
##
## Alternative Hypothesis:    True variance is not equal to 6.25e+08
##
## Test Name:                Chi-Squared Test on Variance
##
## Estimated Parameter(s):    variance = 451980006
##
## Data:                     xiaomi$Price
##
## Test Statistic:           Chi-Squared = 28.92672
##
## Test Statistic Parameter:  df = 40
##
## P-value:                  0.1940353
##
## 95% Confidence Interval:    LCL = 304662625
##                             UCL = 739948892
```

```
sd(xiaomi$Price)
```

```
## [1] 21259.82
```

```
varTest(xiaomi$Price, conf.level=0.99, sigma.squared=sugma, alternative="greater")
```

```
##
```

```
## Results of Hypothesis Test
## -----
##
## Null Hypothesis:                variance = 6.25e+08
##
## Alternative Hypothesis:         True variance is greater than 6.25e+08
##
## Test Name:                     Chi-Squared Test on Variance
##
## Estimated Parameter(s):        variance = 451980006
##
## Data:                          xiaomi$Price
##
## Test Statistic:                Chi-Squared = 28.92672
##
## Test Statistic Parameter:      df = 40
##
## P-value:                       0.9029824
##
## 99% Confidence Interval:        LCL = 283859166
##                                UCL =      Inf
```

```
varTest(xiaomi$Price, conf.level=0.99, sigma.squared=sigma, alternative="less")
```

```
##
## Results of Hypothesis Test
## -----
##
## Null Hypothesis:                variance = 6.25e+08
##
## Alternative Hypothesis:         True variance is less than 6.25e+08
##
## Test Name:                     Chi-Squared Test on Variance
##
## Estimated Parameter(s):        variance = 451980006
##
## Data:                          xiaomi$Price
##
## Test Statistic:                Chi-Squared = 28.92672
##
## Test Statistic Parameter:      df = 40
##
## P-value:                       0.09701764
##
## 99% Confidence Interval:        LCL =      0
##                                UCL = 815691534
```

```
sqrt(815691534)
```

```
## [1] 28560.31
```

```
sqrt(283859166)
```

```
## [1] 16848.12
```

10. Determine a 99% confidence interval for the population proportion of Apple devices, assuming that a representative sample of the smartphone market is available. Test the null

hypothesis that selecting a Apple device has a population probability greater than 0.25.

Apple has 34 individuals, which is almost 4.4% of the population. The 99 confidence interval using a proportion test gives us the interval (2.97%, 6.9%). Performing 2 one-sided tests we see that we can reject that the proportion of apple is equal to 25%, and then we know by its alterantive hypothesis that this sample is below and the population is extremely likely.

H0: $\text{prop}(\text{apple}) = 0.25$ H1: $\text{prop}(\text{apple}) > 0.25$

```
tt <- table(df$company)
tt
```

```
##
##   Alcatel      Apple   Blacear BlackZone   Easyfone      Gee   Hicell      Honor
##       7       34       17       6       19       7       6       12
##   Ikall   InFocus     Inovu     iQ00     Itel     JIVI     Jmax   Karbonn
##      61      14       8       8      22      28       6      39
## Kechaoda     Lava      LG      Mi      Micax  Motorola     MTR     Nokia
##      49      41      11      8      42       5      16      34
## OnePlus     OPPO     Realme   Salora   Samsung   Snexian     Ssky     Vivo
##      19      47      16       7      68      16       7      40
##   Xiaomi
##      41
```

```
prop.table(tt)
```

```
##
##   Alcatel      Apple   Blacear   BlackZone   Easyfone      Gee
## 0.009198423 0.044678055 0.022339028 0.007884363 0.024967148 0.009198423
##   Hicell      Honor     Ikall     InFocus     Inovu     iQ00
## 0.007884363 0.015768725 0.080157687 0.018396846 0.010512484 0.010512484
##   Itel      JIVI      Jmax     Karbonn   Kechaoda     Lava
## 0.028909330 0.036793693 0.007884363 0.051248357 0.064388962 0.053876478
##   LG      Mi      Micax   Motorola     MTR     Nokia
## 0.014454665 0.010512484 0.055190539 0.006570302 0.021024967 0.044678055
##   OnePlus     OPPO     Realme   Salora   Samsung   Snexian
## 0.024967148 0.061760841 0.021024967 0.009198423 0.089356110 0.021024967
##   Ssky      Vivo     Xiaomi
## 0.009198423 0.052562418 0.053876478
```

```
prop.test(x=c(34), n=c(761), p=0.25, conf.level=0.99, alternative="greater")
```

```
##
## 1-sample proportions test with continuity correction
##
## data:  c(34) out of c(761), null probability 0.25
## X-squared = 170.01, df = 1, p-value = 1
## alternative hypothesis: true p is greater than 0.25
## 99 percent confidence interval:
##  0.02970429 1.00000000
## sample estimates:
##           p
## 0.04467806
```

```
prop.test(x=c(34), n=c(761), p=0.25, conf.level=0.99, alternative="less")
```

```
##
## 1-sample proportions test with continuity correction
```

```
##
## data:  c(34) out of c(761), null probability 0.25
## X-squared = 170.01, df = 1, p-value < 2.2e-16
## alternative hypothesis: true p is less than 0.25
## 99 percent confidence interval:
##  0.0000000 0.0663198
## sample estimates:
##           p
## 0.04467806
```

```
?prop.test
```

11. Test the null hypothesis that the proportion of Apple devices and the population proportion of Xiaomi devices is the same at 1% significance level.

We can clearly see that at a any significance level, proportions are not the same

```
prop.test(x=c(34, 41), n=c(761, 761), p=c(0.5, 0.5), conf.level = 0.99)
```

```
##
## 2-sample test for given proportions with continuity correction
##
## data:  c(34, 41) out of c(761, 761), null probabilities c(0.5, 0.5)
## X-squared = 1233.3, df = 2, p-value < 2.2e-16
## alternative hypothesis: two.sided
## null values:
## prop 1 prop 2
##    0.5    0.5
## sample estimates:
##    prop 1    prop 2
## 0.04467806 0.05387648
```

Do not forget to knit your .Rmd file to .pdf (or to word and afterwards to pdf) before posting it on the ATENEA platform task (only for pdf). Markdown should be also posted in the corresponding task.